

Translating Chest X-Ray Explainability into Uncertainty-Aware Radiology Reports

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Abstract

Deep learning models achieve strong predictive performance in chest X-ray interpretation, but their clinical utility is limited by opaque decision-making, unreliable spatial localization, and a tendency for generative language models to hallucinate unsupported medical claims. While post-hoc explainability (XAI) methods like saliency maps exist, they are often spatially unstable, poorly calibrated, and evaluated on pixel-level metrics that do not align with regional radiological reasoning.

This thesis presents an end-to-end, explainability-driven framework that elevates XAI from a qualitative visualization into a calibrated, decision-aware engine for safe radiology report generation. A Vision Transformer classifier is combined with multiple post-hoc explanation methods (Grad-CAM++, GradientSHAP, and Occlusion). To align with clinical practice, these pixel-level signals are aggregated into a 6-zone anatomical lung representation. We introduce a novel, pathology-aware squared fusion strategy that exponentially rewards the most reliable saliency methods. Using only the Top-1 predicted zone, this approach retrieves approximately 45% of the total ground-truth abnormality mass—successfully capturing roughly 80% of the absolute maximum achievable limit for a single anatomical region, whereas, using the top-2 predicted zones retrieve approximately 71% of the total abnormality, capturing approximately 83% of the absolute maximum achievable limit.

To bridge visual evidence with safe text generation, the framework stratifies calibrated classifier probabilities into discrete clinical decision zones (Rule-out, Indeterminate, Suggestive, Rule-in), validated via population-level likelihood ratios (LR+). These zones act as strict constraints for a Large Language Model (LLM), enforcing silence on negative findings and dictating appropriately hedged vocabulary for uncertain cases.

Evaluated against radiologist annotations in the CheXlocalize and CheXpert datasets, the framework demonstrates strong spatial grounding and clinical safety. Crucially, report-level evaluation via CheXbert reveals that the system effectively prioritizes safety over narrative completeness; a majority of recorded False Positives are shown to be metric artifacts where CheXbert rigidly misclassifies the framework's appropriately hedged, Zone 1a uncertainty language as hard positive assertions. However, a subsequent clinical sensitivity analysis demonstrates that treating these indeterminate predictions as diagnostic safety-nets recovers high effective sensitivity and successfully mitigates automation bias. Overall, this work provides a principled pathway toward trustworthy diagnostic AI by strictly linking calibrated predictions, anatomically grounded evidence, and constrained clinical language.

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Chapter 1

Introduction

1.1 Motivation

One of the most popular diagnostic modalities is Chest X-ray (CXR) imaging where it is used for screening and triage. Advancements in deep learning made these applications possible where is has been shown that they achieve high performance when applied on large datasets, and in many cases exceeding radiologists accuracy. For those reasons, those type of artificial neural networks are been explored for decision-support purposes in clinical radiology settings.

Despite these advances, most deep learning models for chest X-ray interpretation operate as opaque black boxes, as they cannot justify or reason on how they come up with the diagnosis. This is an important limitation because, in clinical settings, safety and accountability is of high importance. Radiologists reason by localizing abnormalities, integrating contextual information and communicate their findings carefully by using proper diagnostic language. AI systems fail to do this since its outputs are raw probabilities.

A promising approach to overcome these limitations is called Explainable Artificial Intelligence (XAI), which consists of a set of techniques that attempt to highlight specific regions in images which contribute to the predictions of a deep learning model. Those techniques are called saliency-based and some popular ones are Grad-CAM++, GradientShap and occlusion sensitivity. However, the existing explainable techniques suffer from instability and it has been proven that they have limited localization accuracy. Additionally, the most commonly selected localization evaluation metrics that are used, they are based on pixel-level metrics, which fails to correspond with how radiologists reason regionally and anatomically (e.g., localizing disease to the 'right lower lung zone').

Concurrently, recent progress in large language models (LLMs) has allowed radiology reports to be automatically generated from medical images, as these models are very fluent at writing, in every domain. However, they may usually include hallucinated findings or even produce overconfident language, if the prompt template is not structurally constrained.

The motivation of this thesis is to construct an integrated framework that bridges deep learning with explainability to produce clinically faithful radiology reports.

1.2 Limitations of Existing Approaches

Many automated chest x-ray interpretation systems generate radiology reports by formulating the task as a classification problem, where the predicted probability for each image is used as evidence. While this approach may be useful for triage, those outputs do not include anatomical context, or any spatial information for the abnormality, which radiologists are based on when writing the diagnostic impressions

Saliency-based explainability methods are usually evaluated using pixel-level metrics, such as IoU(intersection over union) or pointing game scores. To some extent, these approaches can produce visually convincing explanations; however, these saliency maps do not align with expert-defined areas of interest.

Furthermore, current radiology report generation systems commonly produce text derived from evidence captured from an image. A common weakness of this approach is hallucinating abnormalities and false overconfident use of language, which makes these reports clinically unreliable.

1.3 Core Idea and Thesis Hypothesis

This thesis proposes that clinically meaningful explainability and radiology report generation require an intermediate, decision-aware representation that explicitly links model predictions to spatial image evidence, uncertainty levels, and reporting constraints.

The central hypothesis of this work is that a chest X-ray interpretation pipeline which: (i) combines multiple complementary saliency-based methods to reduce individual method bias, (ii) aggregates pixel-level explanations into anatomically defined lung regions, (iii) stratifies model confidence into calibrated decision zones aligned with radiological reporting practice, and (iv) constrains language generation using these decision zones and spatial evidence, can generate explanations and radiology reports that are both quantitatively validated and clinically interpretable.

Rather than treating explainability, uncertainty estimation, and report generation as independent or loosely coupled tasks, this thesis unifies them within a single end-to-end framework in which explainability outputs directly inform decision-making and language generation.

1.4 Overview of the Proposed Framework

The proposed system consists of the following components:

1. Multi-label chest X-ray classification with a Vision Transformer model trained on a dataset.
2. Generation of saliency maps using several different explainability techniques.
3. Anatomical partitioning of the image in 6 zones, to map pixel-level saliency into clinically meaningful lung regions.
4. Aggregation of the saliency signals from all saliency methods into zone-level evidence.
5. Decision-zone stratification, assigning each pathology to one of four clinically motivated zones, based on its calibrated output probability :
   - Zone 0: confidently negative (silence)
   - Zone 1a: indeterminate (triggers hedged uncertainty)
   - Zone 1b: probably positive
   - Zone 2: rule-in positive
6. Likelihood-ratio calibration grounding decision zones in population-level performance statistics.
7. Constrained report generation using structured evidence into an LLM prompt template.
8. Evaluation using both spatial localization metrics and radiology report analysis.

This modular design allows each component to be independently evaluated while contributing to a coherent clinical narrative.

1.5 Clinical Motivation for Decision Zones

Radiological interpretation is inherently graded rather than binary. In clinical practice, findings are routinely categorized as absent, equivocal, suggestive, or definitively present, with reporting language carefully adjusted to reflect diagnostic confidence. Conventional binary classifiers fail to preserve these distinctions, forcing premature commitment and obscuring uncertainty that is clinically meaningful.

This thesis introduces a decision-zone framework that explicitly models this graded reasoning. Rather than directly mapping model probabilities to textual assertions, predictions are stratified into four decision zones derived from population-level performance characteristics: a negative zone (Zone 0), an uncertainty zone (Zone 1a), a weakly positive zone (Zone 1b), and a rule-in zone (Zone 2).

Decision-zone boundaries are calibrated using sensitivity-anchored thresholds, sensitivity–specificity intersection points, and high-specificity rule-in criteria. This design ensures that uncertainty is not treated as noise but as a first-class output of the system. Zone 0 findings are suppressed entirely, Zone 1a findings are reported with explicit uncertainty, and progressively stronger diagnostic language is permitted only in higher-confidence zones.

By aligning model behavior with established radiological reporting practices, the proposed framework enables strict silence when evidence is insufficient, transparent communication of uncertainty, and assertive reporting only when empirically justified.

1.6 Contributions

The primary contributions of this thesis are as follows:

- A multi-method explainability fusion framework that integrates Grad-CAM++, GradSHAP, and Occlusion saliency into pathology-specific, anatomically grounded lung-zone evidence.
- A novel pathology-aware squared fusion strategy ($MeanTop - 1^2$) that exponentially rewards highly reliable saliency methods, significantly outperforming individual and simple-average XAI techniques in ground-truth mass retrieval.
- A decision-zone modeling approach that stratifies predictions into negative, uncertain, suggestive, and rule-in categories using empirically calibrated sensitivity, specificity, and likelihood-ratio criteria.
- A decision-aware radiology report generation pipeline that enforces evidence-based reporting, suppresses unsupported findings, and constrains diagnostic language according to calibrated uncertainty.
- A comprehensive evaluation protocol combining spatial localization analysis with report-level clinical NLP assessment using established radiology benchmarks.
- An end-to-end system architecture that unifies explainable AI, uncertainty modeling, and constrained natural-language generation into a clinically interpretable diagnostic framework.

1.7 Thesis Structure

This thesis is organized to reflect the progressive construction and evaluation of a decision-aware chest X-ray interpretation system.

Chapter 2 reviews related work in chest X-ray classification, explainability, localization, and radiology report generation, highlighting limitations in existing

evaluation and reporting paradigms. Chapter 3 describes the datasets, preprocessing steps, and experimental splits used throughout the study. Chapter 4 presents information about how the classifier was trained and what techiniques were employed.

Chapter 5 details the explainability fusion and anatomical localization methodology, including the construction of zone-level evidence. It also introduces the decision-zone framework and its calibration using population-level performance and likelihood ratios. Additionally, it describes the constrained radiology report generation approach, emphasizing uncertainty-aware and evidence-grounded language control.

Chapter 6 presents experimental results across localization and report-generation tasks. Chapter 7 discusses the findings and the limitations of the proposed framework. It also concludes the thesis and outlines directions for future work.

Chapter 2

Related Work

2.1 Chest X-Ray Classification

Deep learning has been very successful in the task of chest radiographic classification, as it has been shown in experimental results in large datasets such as NIH ChestX-ray14, CheXpert, MIMIC-CXR, and many more[1]. Moden neural network architectures like CNNs can achieve expert-level sensitivity and specificity in the detection of common thoracic pathologies. Early work by Rajpurkar on pneumonia diagnosis reported high performance comparable to radiologists, and several models on the CheXpert dataset even surpassed radiologist accuracy in a lot of pathologies such as cardiomegaly, edema, pleural effusion[2]. Vision Transformers(ViT) have also been explored as an alternative. For example, park et al. employed a ViT to classify radiographs achieving good results on data from three hospitals[3]. It is apparent from all these advances that deep learning is a very promising approach in chest radiography.

Even though these deep learning models can achieve high ROC scores, they face limitations in clinical contexts because they oversimplify the findings by either classifying them as positive or negative. A real radiology report for example, contains information about the location of an abnormality, its extent and commonly noting the uncertainty. In contrast, deep learning models just output a confidence score(e.g pneumonia: 90%) without any other details regarding the anatomical location, which is a critical limitation for clinical utility and reliability.

Additionally, it is challenging to handle the uncertainty in chest x-ray classification. Radiologists commonly use phrases like "possible pneumonia" or "cannot exclude nodules" to reflect the uncertainty since the findings may not always be clear. The CheXpert dataset included the uncertainty in the ground truth labels where a case may be either positive, negative, or uncertain. Studies also shown that the performance of a model may differ depending on how one decides to treats the uncertain cases during training[2]. Even though there is not any single best practice, ignoring or relabelling the uncertain cases during training often

leads to models that are poorly calibrated to true likelihood, and this is critical in medical AI and in decision making because it may mislead clinicians[4].

Another important concern has to do with generalization and bias. It has been observed that many classifiers may learn false correlation from the Chest x-ray metadata. For example, a model may correlate an AP chest X-ray label with ilness or that model have been focused their attention on hospital-specific markers or other metadata artifacts that are overlaid in the radiograph, instead of focusing on true pathology features[5]. This can cause failures when the model is tested on slightly different data, which makes a strong argument that interpretability should be considered even more, beyond just accuracy metrics.

To summarize, it is a fact that CNN and ViT clasifiers are very effective chest x-ray classifiers but their outputs do not include any reasoning of the diagnoses and this causes a misalignment with the clinical needs, as clinicians the ability of localizing findings and conveying uncertainty as very important characteristics. Additionally, the pressence of label noise, bad calibration and model bias add barriers for clinical use. All these limitations motivate approaches for the explainability integration, calibrated confidence and outputs that are more qualitative as radiology reports are.

2.2 Explainability Methods in Medical Imaging

Many attempts have been made to tackle the "black-box" nature of the deep learning models. The most common approaches include saliency-based or pixel-attribution methods. Gradient-based techniques like Gradient-weighted Class Activation Mapping (Grad-CAM) and its variants (Grad-CAM++), backpropagation-based attributions (e.g. Integrated Gradients, DeepLIFT, Gradient $\times$ Input, Guided Backprop), and perturbation-based methods such as occlusion maps[5]. All these methods essentially produce a heatmap that is overlaid on the image highlighting regions where the model focused on to make the prediction, and these methods are popular as they do not require any annotations during training. For an instance, Grad-CAM can be applied to a classifier trained on chest radiographs and it can produce a heatmap over the lungs to visually justify where it focuses to determine that the condition is present. This visual explanations are presented to clinicians as a proof that the model is "looking" at clinically relevant regions and they are not randomly making diagnoses.

However, there are some concerns regarding the utilization and reliability of saliency maps in medical imaging, as it has been observed that some saliency maps are insensitive to model parameters which means that a model with random weights may still get similar visual explanations to a trained model[6]. Another concern is raised from the case where small input changes or adversarial noise may significantly alter the highlighted regions, which makes these techniques be unstable. It has also been reported that saliency methods may in some cases highlight irrelevant areas, such as the patient ID label, or other markers. In essence, these post hoc techniques provide some visual justifications but they are not guaranteed to be correlated with the model's true reasoning

or the ground truth pathology locations.

Additionally, another challenge in saliency methods is the evaluation step. Saporta implemented a benchmark of seven saliency methods(including Grad-CAM, Integrated Gradients, etc.) on chest radiographs including radiologist annotated ground truth segments and they observed that all saliency methods perform worse than a radiologist, and in some cases, by a wide margin[5]. Even though Grad-CAM was one of the best methods performance-wise, it still performed much worse on localization overlap with ground truth compared to humans and this difference was more clear on small or complex shapes of the abnormalities. This suggests that a model with high accuracy usually fail to visually highlight clinically relevant regions. Another observation concluded that a model that has made a prediction with high confidence tends to have better saliency accuracy, while, predictions that were uncertain often produced diffuse visual explanations. The community generally uses simple metrics and benchmarks to compare different methods and they will be discussed in the next section.

Furthermore, these saliency explainability tools are post-hoc and are not integrated in the model's decision making which means that they are used on top of the trained model, and this separation can sometimes cause misalignments with the actual prediction. For example, a classifier might correctly output a high probability for Pneumonia but the Grad-CAM heatmap highlight a wrong region. This misalignment may lead the user to distrust a correct diagnosis. The inverse scenario can also occur, where a heatmap coincidentially overlap with a clinically relevant region while the classifier made a negative diagnosis. This may lead the user to incorrectly trust the visual explanation, so there is a risk of overestimating these visual evidences and this confusion may affects clinicians decision making[7].

In summary, post-hoc methods such as Grad-CAM, Integrated Gradients, occlusion sensitivity, and others are useful tools on visualizing CNN models and they have been commonly applied in chest X-ray diagnostics. They have discovered cases where models were focusing on irrelevant regions but they generally offer a level of transparency on model's predictions that are rather opaque. Their limitations are potential inaccuracies, instability, and misalignment with the model decision process. These findings motivate researchers for more robust and integrated explainability in medical imaging, a domain that is critically important. We will discuss recent works that do not rely on a single saliency method but have explored combining explanations or using better evaluation frameworks.

2.3 Localization and Evaluation of Explanations

An important characteristic in explainable medical imaging is localization, that is, inspecting the location where the model considers that an abnormality is on the image. If a model activates the correct region of a pathology, then, this explanation is consider good. This type of evaluation is possible through datasets

that include ground-truth localization. One such example is the CheXlocalize dataset introduced by Saporta[5], which included radiologist-annotated polygons. CheXlocalize essentially provide pixel-level segmentation masks for 10 thoracic pathologies(e.g pleural effusion, cardiomegaly, edema, etc), on a set of images from the CheXpert dataset. More specifically, it includes two separate sets, a validation set containing 234 images and a test set, containing 668 test images with polygons drawn by board-certified radiologists, and among those, the dataset also includes annotations that shows the location of the most representative location of each pathology. The release of ChwXlocalize motivated researchers for quantitative studies of explanation accuracy in chest X-rays.

Another example of such dataset is called VinDr-CXR[1], from Vietnam, which provides a large set of 18,000 chest radiographs including radiologist-drawn bounding boxes for 22 pathologies. Probably the largest publicly available chest X-ray dataset with expert localization annotations per image and labels on a test set. This dataset started by an Kaggle competition on abnormality detection with the emphasis on the importance of localization evaluation. Other earlier but popular datasets that can be mentioned are the NIH ChestX-ray14 which included a small set of images with bounding box annotations, and also, the PadChest dataset where it provided segmentation for specific pathologies but these were more limited in richness. So, through these relatively recent datasets, the community could conduct studies to evaluate a variety of saliency maps. Some of the quantitative metrics that are used in the literature to evaluate explanations are the following:

IoU (Intersection over Union) or related overlap scores (Dice coefficient, etc.): They treat the saliency map as a segmantation. By applying thresholds on the values of the saliency map to derive a region and computing the overlap with the ground-truth segmentation mask[8]. In the Saporta et als's study, the mean IoU(mIoI) over a test set is used as the primary metric. A high IoU is translated as having an explanation that overlaps a lot with the radiologist mask. However, IoU is too strict: it requires the saliency to cover the full extent of the lesion to score highly. If a heatmap only highlights part of a large abnormality, IoU will penalize it even if that part is indeed within the abnormal region.

Pointing Game / Hit Rate: This metric focuses only on the most high value in the explanation. It checks if the highest-valued pixel from the saliency map is located inside the ground-truth bounding box or mask of the abnormality. If yes, it counts as true, else, it counts as false. Then it calculates the percentage of images for which the explanation correctly were included inside the true region[9]. It is a looser criterion compared to IoU since it only requires for the explanation's peak to be on the target area. In some clinical applications, the correct prediction of the targeted area may be sufficient, rather than predicting the full shape.

Region-based or zone-based accuracy: Rather than demanding exact pixel-level overlap, another strategy is to project both the explanation and the ground truth onto a set of predefined anatomical zones (e.g., left upper lung, right lower lung, mediastinum) and evaluate their agreement at that zonal level.

This coarser assessment recognizes that perfect shape correspondence may be less important than correctly localizing the finding to the appropriate anatomical region[10]. For example, emphasizing the left lung base in a case of left lower lobe pneumonia could be considered adequate at the zone level, even if the saliency map does not perfectly trace the margins of the consolidation.

Although this idea is intuitively appealing from a clinical perspective, zone-based evaluation has seen limited uptake in the existing literature. Benchmarking efforts predominantly emphasize pixel-level overlap and pointing game metrics. A few earlier studies have examined correctness at the lung or quadrant level in scenarios with only coarse localization labels, but these approaches lack standardization and are not tied to confidence estimation. This thesis proposes a systematic zone-based evaluation framework that unifies explainability and decision-making, thereby filling an important gap in radiology report generation.

The limitations of these metrics should be noted. Metrics like IoU and Dice depend on the thresholding of the saliency map and on the ground truth mask. Also, small spatial differences in extent can significantly reduce IoU results even though the saliency can be sufficient. The pointing game metric ignores the extent of the localization in its entirety which may not be considered as a complete explanation. Usually though, multiple evaluation metric are employed. Saporta observed that saliency methods on average achieved only 24% overlap with the ground truth annotations compared to human benchmarks, and they also observed worse performance in the hit rate as well[5]. So it has been established that the algorithm localization and expert localization differ.

Another important finding from the analysis in Saporta's paper derived through the failure case analysis on the explanations. One example had a high IoU but considered a "miss" because the peak value of the saliency was not located inside the lesion, whereas, in another case that had a very low IoU was a "hit" because the saliency max value landed inside a very small ground truth area. This analysis suggest that using a combination of metrics is usually better.

Finally, while most saliency evaluations in literature related to pixel-level overlap against bounding boxes or segmentation maks, they are not the most appropriate ones for radiology report generation. A zone-level (e.g., upper/mid/lower lung $\times$ left/right lung) metric system could however be an interesting approach. This motivates evaluations that summarize both ground truth and model explanations as distributions over anatomical zones by dividing the lung regions in six segments, enabling new metrics.

To summarize, the publicly available datasets of CheXlocalize and VinDr-CXR allowed for benchmarking of explainability in chest radiographic classifiers. Common evaluation metrics such as IoU and pointing game are used to measure the alignment of saliency maps and expert annotated masks. Studies also have shown how using only one evaluation metric is not a sufficient way to capture the explanation in its entirety, and there is a general motivation for the development of better explainable techniques with probably integrating them into the model training as well.

2.4 Fusion of Explainability Methods

As it has been observed that each explainable method has its own strengths and weaknesses, studies has also explored an approach of combining multiple saliency maps to yield a more robust outcome, with an assumption that the results will be improved as it happens with ensemble learning that improves accuracy.

One such approach combines saliency maps from multiple models or multiple explainable methods to produce an ensemble heatmap. A work on thoracic disease localization conducted by Aasem and Iqbal[11], where they trained many CNN classifiers and fused their Grad-CAM class activation maps. The new heatmap which is an aggregation from the result of multiple classifiers aimed at reducing noise and preserve the correctly ground truth regions. They indeed noticed an improvement of the abnormality localization on the IoU metric, in comparison with using a single model's CAM. The reason behind it is that this ensemble approach reduces variance where the values are averaged and the noise may be corrected by another model's CAM. They also employed tranfer learning on the Ensemble-CAM and trained their models on many datasets to increase generalization and mitigate model variability issues.

A different approach that has been explored was to combine explainable methods. For example, the output of an occlusion map, a Grad-CAM and Integrated gradients can be combined either by weighting or using a logical operator(AND/OR). The reasoning is that it increases our confidence that a specific region is actually important when multiple diverse explainable methods highlight the same area. One such study outside of medical imaging was conducted by Dwivedi[12], for face recognition where they integrated saliency maps from Grad-CAM, LIME, and DeepLIFT to construct an improved visual explanation. Other approaches related to the medical domain explored the averaging of different saliency maps, or testing a voting scheme where pixels or regions are highlighted only if the majority of the explainable methods agree on. There is not a single best practice currently favored however.

It is challenging to combine different saliency methods because their outputs may have different scale and meaning. For example, the outputs of Grad-CAM considers only positive attribution, whereas occlusion outputs can have both increases or decreases in output, so they may not be directly comparable. Additionally, the fusion may worsen outcomes if the model's decision itself is weak. Another attempt is to calibrate the fused saliency maps on hit/miss or IoU localization. A recent work also explored associating saliency maps with the model's predicted probability, where visual explanations are important only when the model is confident.

An interesting direction by Salvi(2025) is integrating uncertainty quantification into the visual explanations, stating that this merge can lead to more faithful interpretability[13]. In their proposition, they utilize the model's uncertainty to modulate explanations by shading the heatmap differently in the cases where the model has low predicted probabilities, or highlighting only features that consistently appear across ensemble samples. They justify this framework

by stating that the model's confidence score should be aligned with the generated heatmaps so that ambiguity can be also expressed on them.

Until now, most fusion frameworks are just heuristic attempts and there is not any well defined weighting scheme in theory. Furthermore, there is not a single ideal visual explanation other than the comparison with ground truth radiologist annotations that even themselves are not fully overlapping. It is certain though that the explainability from single saliency maps are unstable which motivates for ensemble approaches to be more considered.

Another approach that has been explored that is related to this subsection is the combination of multi-modal explanations. It involves associating images with generated texts. This idea is increasingly studied in radiology with the development of vision–language models that use text-guided interpretability during training.

To summarize, there is interest in the fusion of explainable methods but more work needs to be done. Approaches that construct Ensemble-CAM by aggregating CAM from multiple classifiers can improve localization. The combination of multiple saliency maps are as well a promising approach to yield more robust visual explanations even though some challenges arise. Ultimately, the aim should be to have an explainable framework that is faithful, can utilize all available information and can most importantly support decision making. This thesis aims to move in this direction for the purpose of generating decision-aware radiology reports.

2.5 Radiology Report Generation

The task of disease diagnosis from classification models is only one of the things that radiologists also do. Radiologists ultimately write medical reports where they describe details about the findings, their impression and other clinically related information. This motivated researchers to automate the generation of radiology reports instead of just training a deep learning model to predict a label. So the aim is to output a paragraph that resembles a medical report, given a chest radiograph.

Some early works approached this problem by using a CNN-RNN architecture, where a CNN would encode the image, and the RNN networks would the medical report sequentially word by word or sentence by sentence. Specifically, Jing(2018) created his own model incorporating an attention mechanism. In this architecture, the LSTM decoder could focus simultaneously at different locations of the image and generate different sentences producing eventually a report[14]. Another alternative approach employed hierarchical models which generates a report as multiple sentences. Some other example, Xue (2018)[15], and Yuan(2019)[16], also explored attention and hierarchical decoding to generate coherent radiology reports. These studies demonstrated the capabilities from the CNN-RNN architectures on datasets such as the IU X-Ray and the MIMIC-CXR. Some of the evaluation metrics that they had used are BLEU,ROUGE and METEOR which are NLP based.

Lately, some other architectures that have been explored for the task of medical report generation are transformer-based. Models like R2Gen and M2TRANS use an image encoder (which could be a CNN or a Vision Transformer) and a transformer decoder for text[17]. It is important to also mention the emergence of Vision-language models, that is, a ViT or Swin Transformer used as the visual backbone paired with a pretrained language model (like GPT-2, BART, or a specialized medical language model) as the text generator. Islam et al(2025) evaluated combinations such as ViT-B16 + GPT-2 and Swin + BART on report generation [18]. These transformer approaches often perform better at the BLEU/ROUGE metrics compared to the older CNN-LSTM models, due to better modeling of long-range dependencies and the utilization of large-scale language pretraining. It has also been observed upon the IU X-Ray dataset, that a Swin-BART model produced more fluent medical reports compared to an LSTM-based model.

However, the task of automating report generation is still challenging. The main issues are documented below:

Hallucinations: Models tend to inlude content that should not be derived from an image. For example, the model may falsely mention a finding or make a comparison to a previous study without actually having context of the patient's history(e.g model may say "compared to previous study..."). Several methods have been proposed to tackle this problem. In another instance, it may hallucinate measurements(e.g "heart size is borderline enlarged"). In essence, these models can actually produce fluent plausible statements, but they may be false. In an error analysis work conducted by Yu et al. (2023)[19], the most significant error they came across are false positives, where the generated text included diseases that were absent from the the ground truth radiology report.

Exclusion of findings: not mentioning a finding that is present in the image. The generated report might miss to mention an abnormality that actually exists in the ground truth radiology report, which makes it a false negative error, a very critical one. One of the reasons this is happening is because of model bias where they tend to report the most common observations[20].

Wrong attributes: The finding can be mentioned but describe the wrong location or severity. For instance, it might mention that an abnormality is located on the left lung while it actually is present on the right lung, or it may mention that the abnormality is large when in reality it may be mild. This means that the model can't comprehend fully the spatial information so it cannot quantify it properly. One other observation is that the model may use common severity words(e.g moderate pneumonia) that does not reflect the image necessarily.

Overconfident language: Radiologists commonly use a diverse vocabulary to implicitely state high confidence or umbiguity(e.g. "consistent with pneumonia, though clinical correlation advised", but ai-generated reports often struggle to do this. For example, a synthetic radiology report may mention "No atelectasis" definitively even in a scenario where the image is of poor quality but a radiologist would be more descriptive and say "No visible atelectasis, though a small atelectasis cannot be excluded given low lung volume.". This is caused partly

as a result of maximizing likelihood of the ground truth text during training. Another similar problem from the synthetic reports is that they are not very diverse in a way that a human would use words to give emphasis when it is needed.

Fluency and coherency issues: Even though transformers have been improved this, they may mix the structure(e.g., mixing findings in the impression section or vice versa). Also, in cases where the training data have bias or are limited, the model may not learn correctly the medical termnilology or may use it wrongly. They generally also struggle to understand this vocabulary and anatomical structures which leads poor text that does not look proffesional or clear[18].

These challenges suggest that the NLP metrics that are used to evaluate the generated reports are not sufficient, as metrics such as BLEU and ROUGE do not correlate well with clinical accuracy. A high overlapping of words between the reference report and the synthetic report yields a good BLEU score but still, the model may omit crucial findings. One alternative metric emerged from CheXbert: an automatic labeler that uses a BERT model to extract the presence/absence of 14 pathologies given a radiology report as input. More specifically, CheXbert takes as input both the generated report and the reference report, then outputs a vector for each input, so one can then compute a label vector similarity to verify the pathologies that the synthetic report mentioned correctly[21]. However, CheXpert labels are limited to 14 abnormalities and don't capture finer details.

Another metric that is used is called RadGraph. RadGraph is an information extraction schema that takes a radiology report as input and produces a graph of clinical entities(findings, anatomical locations) and their relations. Joshi et al. (2022) introduced RadGraph and an F1 metric to calculate the overlap of these entities/relations between synthetic and reference reports[22]. The rationale behind it is that synthetic report is considered clinically correct if it contains the same anatomical characteristics and findings(for example, mentioning "moderate pleural effusion in the right lung base" just as the reference does). It has been found that RadGraph F1 metric have a stronger correlation with how radiologists evaluate a report in terms of correctness, compared to the BLEU or CheXbert metric.

A study conducted by Yu et al. on radiology report evaluation propose a new metric named Rad-CliQ (Radiology Clinic Score), which is a combination of Rad-Graph, CheXbert similarity, and text similarity, aiming to predict the number of errors a report has[20]. They observed a strong correlation of Rad-CliQ and RadGraph F1 with radiologist rationale, while BLEU had the worst correlation confirming the limitations of using naive metrics. They also concluded that missing to mention a pathology is far worse than grammatical error or poor fluency. However, researchers still work on for metrics that explicitly measure clinical factuality of medical reports.

To summarize, the automation of radiology reports started from CNN-RNN architectures and progressed to transformer and vision-language models. The generated reports now look more fluent and usually manage to capture the

essence of a radiograph. However, errors occur such as hallucination of findings, the missing of abnormalities, and struggles with uncertainty, which make them generally not reliable enough. Evaluation metrics such as CheXbert and RadGraph emerged that better reflect clinical correctness. However, current models are not perfect and progress needed to be done to face these errors. We can essentially observe that these model that generated the reports are not really know why they are saying something about the image. This motivates a new framework which includes better explainability and decision modeling into the generation process, allowing a model's generated text to be grounded in an interpretable representation (e.g. detected findings with locations and uncertainties). The next section is concerned on how uncertainty and decision modeling can be connected with both classification and generation tasks in the clinical context.

2.6 Uncertainty, Confidence, and Clinical Decision Modeling

Uncertainty is often present in clinical practice and radiologists should manage it properly. A radiologist may communicate an ambiguity by suggesting a follow-up imaging or use different phrases to state degrees of certainty, such as "suggestive of" or "consistent with". It is important that the outputs of the models that make diagnosis be aligned with how clinicians handle uncertainty for safety reasons.

The way that deep learning models make predictions is by generating a continuous probability as output. So, given a chest radiograph as input, the classifier may predict the probability that pneumonia is present. Commonly, a threshold is chosen(e.g 0.5) so that the observation can be categorized either as positive or negative. For an instance, the bigger the value is above the threshold, the more confident the prediction is that the pathology is present, and vice versa. In a case where the output probability is let's say 0.51, it is considered an equivocal. It has been argued that, by providing the magnitude of the output score to the user helps to decide if the prediction can be trusted[23].

However, the problem is not solved by just providing the raw model's value. A good practice is to calibrate the model so that its output correspond to the actual probability, so that the model can be more trusted. A system that is proposed by Karunanayake et al. (2025) state that AI systems should provide the model's confidence to the user only for the high-uncertainty cases so that radiologists review them and by doing so, they avoid overloading clinicians with numbers for every case[24]. To implement this, one should tune the appropriate operating points and turn the model to the hospital's population so that, for instance, 95% confidence" really means 95% in that context. In this way, the model can communicate correctly to the clinician when it is uncertain, which helps protect against automation bias.

One more formal approach is to associate the model outputs to clinical deci-

sion theory. For example, bayes' theorem or likelihood ratios can be computed as a post-test probability, given that a patient, a patient let's say has a 10% pre-test probability of disease and and the diagnosis of a chest X-ray is positive with a certain sensitivity/specificity. So, AI outputs can be associated with likelihood ratios, which shows, how much more likely disease is given a positive result. For instance, an AI system might achieve a positive likelihood ratio (+LR) of 11.7 and a negative likelihood ratio (-LR) of 0.18 for a pathology. That means a positive AI prediction increases the odds of disease 11-fold (strong rule-in power), while a negative prediction reduces the odds to 0.18× (strong rule-out power)[25]. Clinically, one could set thresholds such that if the AI is very confident negative (and thus -LR is low enough), the case could be ruled out without a radiologist reading, or moved down the priority list. Conversely, a very confident positive might be fast-tracked for follow-up.

Current studies also focus on the usage of triage and "rule-in/rule-out" tasks. A work conducted by Oakden-Rayner et al. (2020) proposed a frameworks where a high-sensitivity threshold can be selected to capture as many positives as possible and rule-out normal chest radiographs to reduce workload for radiologists[26]. Another study conducted in mamography selected thresholds associated with +LR of 10 as a rule-in decision, and -LR of 0.2 for rule-out, with the goal of aligning the outputs of a classifier with clinically relevant cut-offs[27]. It is important to be noted though that the model should be well-calibrated and uncertainty cases are been handled appropriately.

Some methods that estimate the epistemic uncertainty of models are Monte Carlo dropout, deep ensembles, and Bayesian neural networks, and they have been studied in radiology with the purpose of detecting cases where the prediction of the model may be less reliable. A recent review by Begoli et al. (2020) noted that predicting uncertainty can help flag "at risk of AI failure" cases[28]. For instance, in an MRI cancer diagnosis task, cases with high prediction variability were apparently the ones where the classifier often was wrong. In chest X-rays, an ensemble of models could output not just a single value but an uncertainty interval. If the models disagree widely (say one predicts 0.2 and another 0.8), the case has high epistemic uncertainty which indicates that maybe the appearance is unusual. This is similar to the variability that exists in radiology, where the same radiograph is given to five radiologists and they don't all agree, the case is probably difficult or ambiguous. AI can capture a similar notion by noting when its "committee" of network predictions diverges.

It is complicated to communicate uncertainty to the user but it is potentially useful. Harrison.ai proposed that the output score could be accompanied with an uncertainty range e.g., "Pneumothorax probability 60% ± 20%". For example, a small predicted probability included with a very narrow range informs the user that the model is very certain that there are no findings in the observed case(increased trust)[29]. In another scenario, a high predicted probability included with a wide uncertainty range could prompt the clinician for extra caution so the case will be double-checked or another test could be ordered. Overall, by managing and displaying confidence correctly, users can decide when to trust the AI and when to be skeptical, improving overall diagnostic accuracy

and reducing errors. However, many AI systems that exist currently do not include uncertainty estimates and the end-user is not informed in scenarios where the model might fail.

Finally, it's important to align AI confidence with clinical reporting practices. Radiologists rarely use numerical probabilities in reports (some exceptions exist, like BI-RADS categories in mammography that map to risk percentages, but even there they use categories like "probably benign"). Instead, radiologists use qualitative certainty terms (definite, likely, possible, unlikely, etc.)[30]. There have been efforts to standardize this language – for example, the ACR has guidelines for terms corresponding roughly to probability ranges (e.g., "almost certainly" 99%, "likely" 85%, "could be" 50%, etc.). An AI system that can translate its numeric confidence into such terms or into a recommended action (e.g., "recommend follow-up CT" when uncertainty is high for something potentially serious) would fit more naturally into clinical workflows. One paper in JACR 2025 discusses frameworks for communicating diagnostic certainty, noting that radiologists prefer broad categories over precise percentages. The goal is to convey uncertainty in a way that informs management, not to pinpoint an exact number. For instance, saying "there is a 4% chance of pulmonary embolism" is less actionable than "a small pulmonary embolism cannot be excluded, consider a CT angiogram if clinical suspicion remains" which is how a radiologist might phrase it. AI should ideally do the same. Instead of just giving a 4% probability, it might say "no acute findings, though the confidence is low due to image quality; recommend clinical correlation or additional imaging if needed." This requires the AI to have an intermediate decision model, understanding what level of certainty triggers what advice, much like clinical decision thresholds (like the Wells' criteria for PE where below a certain probability you might not treat, etc.).

In summary, integrating uncertainty and decision modeling into AI for chest X-rays is essential for bridging the gap between algorithm outputs and clinical decisions. Techniques for uncertainty quantification can identify when the model is unsure, and decision-theoretic concepts (likelihood ratios, utility thresholds) can guide how we use model outputs (for triage, rule-in, rule-out). Emerging viewpoints advocate for treating model confidence as a first-class output that is calibrated, contextual, and presented in a user-friendly manner. The current gap in most literature is that explainability and uncertainty are treated separately – saliency maps explain what the model found important, while uncertainty estimates tell us how sure the model is, but rarely are the two combined into a cohesive explanation (e.g., "the model highlights the left lower lobe opacity and is 90% confident it represents pneumonia, but with a 10% uncertainty range due to atypical appearance"). Such a decision-aware, explainability-grounded reporting approach is not yet standard. Addressing this gap is a motivation for this thesis: to create a framework where the model's decisions (including whether to be confident or defer) are intrinsically linked with its explanations and are carried through to the final report output.

2.7 Summary and Positioning of This Thesis

Across the areas reviewed – chest X-ray classification, explainability, localization, fusion methods, report generation, and uncertainty modeling – we observe a common theme: existing solutions address pieces of the puzzle, but none provide a complete, integrated approach for decision-aware, explainable radiology reporting. There are clear gaps in the literature that this thesis aims to fill.

First, while CXR classifiers can detect pathologies with high performance, they operate in isolation from explainability and reporting. They give us probabilities or binary labels, but not the rationales or the nuanced descriptions a radiologist would provide. On the other hand, report generation models produce fluent text but currently function as black boxes that occasionally hallucinate or omit findings, without a robust way to ground each sentence in the image evidence. No existing method fully connects the pixel-level reasoning of a model (e.g. where it is looking and why) with the language output in a rigorous way. Some attempts add attention maps or keywords to justify generated sentences, but these are weakly tied and often not trusted by clinicians.

Second, explainability research has mostly focused on post hoc visualizations or at best on evaluating them, but not on using them to improve decision-making or generation. We lack mechanisms to enforce during model training that an explanation be "attached" to each decision or each sentence in a report. This thesis posits that an intermediate, decision-aware representation can bridge this gap. For example, instead of going directly from image to text, a model could first predict a structured set of findings with associated locations and confidence (an internal "decision report"), and then generate text from that. Such a representation could be inherently explainable (it's basically what a radiologist would jot down as observations) and also guide the language model to produce accurate, evidence-backed sentences. Some recent works hint at this direction: the Chest X-Ray Explainer (ChEX) by Müller et al. (2024) introduces an interactive model that allows querying by region and outputs region-specific descriptions. That shows the value of an architecture where anatomical localization and textual description are jointly modeled. However, even ChEX primarily focuses on interactivity and does not explicitly incorporate uncertainty or a formal decision model.

Third, the literature on uncertainty in medical AI is somewhat separate from the literature on explainability and reporting. This thesis argues these should be combined. If a model is uncertain about a finding, that uncertainty should propagate to the explanation and the report. For instance, instead of the report saying "No pneumothorax," it could say "No definite pneumothorax, but apex not well visualized" if the model's confidence is low and it knows the reason (e.g., part of the lung is obscured). No current end-to-end model does this. Salvi et al. (2025) propose merging XAI and UQ for reliability, and Karunanayake (2025) emphasizes showing confidence only when it alters decisions – both echo the need for decision-aware explanations. Yet, implementations of these ideas in a CXR reporting system have not been seen. Most report generation models do not even output a confidence score for their findings.

Finally, evaluation metrics themselves underscore the gap: RadGraph and RadCliQ evaluate whether the content is correct and traceable to findings. If a model could explicitly output the RadGraph (i.e., a graph of findings and relations) as an intermediate step, we would have a much more transparent system. Existing models do not output structured findings; they jump to text. Thus they fail to "connect the dots" between image evidence → internal reasoning → verbal report. This thesis is positioned to create that missing link by designing a framework where the internal reasoning is explicit, inspectable, and grounded in decisions. In doing so, it will draw upon the prior work in each sub-area: using high-performing classifiers and transformers (from Section 2.1 and 2.5), ensuring explainability and localization of those internal decisions (Section 2.2 and 2.3), possibly combining multiple explanation signals (Section 2.4), and incorporating uncertainty estimation (Section 2.6) into the pipeline.

In summary, the state of the art provides strong individual components – accurate classifiers, informative saliency maps, fluent report generators, uncertainty estimators – but fails to integrate them into a cohesive whole that meets clinical needs. The literature lacks a method that seamlessly links pixel-level model explanations with an uncertainty-calibrated decision process and with the final narrative report. This thesis aims to fill that void. By proposing a decision-aware, explainability-grounded reporting framework, we seek to ensure that what the model outputs (the report) is directly supported by why it outputs it (the explanation), and tempered by how sure it is (the uncertainty). In doing so, we address the crucial gaps identified: bridging explainability and report generation, tying localization to language, and embedding uncertainty into the model's communication. This approach is a natural evolution of the literature towards an AI that not only achieves high performance, but also justifies and qualifies its conclusions in a manner aligned with clinical practice.

2.8 Positioning of This Work

In contrast to prior approaches, this thesis integrates classification, explainability, localization, uncertainty calibration, and report generation into a single coherent framework. Rather than treating explainability as a visualization artifact, it is used as a decision-relevant signal that informs reporting behavior.

By combining multi-method explainability fusion, anatomically grounded zone-level reasoning, calibrated decision zones, and constrained language generation, this work advances toward clinically faithful and trustworthy chest X-ray interpretation systems.

The proposed framework complements existing classification benchmarks while addressing key limitations in interpretability, localization validity, and reporting safety.

Chapter 3

Dataset Description and Preprocessing

3.1 Overview of the CheXpert-small Dataset

The public dataset that was selected for this work is named CheXpert, which consists of a huge number of radiographs and it was created at the Stanford University to allow researchers to train deep learning models that can make diagnoses. A new version of that dataset that was released in 2022 is named CheXpert Plus and it also includes the corresponded radiology reports.

During training, we used all the images of the CheXpert dataset. However, in our explainability analysis and evaluation, which we will later refer to, we only considered the frontal chest x-rays, since they are more common but also because the region-level annotations that were provided by radiologists included exclusively frontal images.

All radiographs are grayscale and differ in resolution. Before we trained the model and computed the saliency maps, we resized the images to a fixed resolution which aligns with the model architecture that we selected, so that the input dimensions were consistent both for the task of classification and the XAI pipeline as well.

3.2 Label Taxonomy and Uncertainty Handling

One of the characteristics of the CheXpert dataset is the methodology of annotating the labels to account for the uncertainty. The labels were annotated from the corresponded radiology reports by using an NLP model which is named Chexbert and was developed by the authors of the dataset. The possible values that Chebert annotated each observation are the following : positive (1), negative (0), uncertain (-1), or missing (NaN).

More specifically, the uncertain labels reflect situations where the associated

language of a radiology report is not confident enough(e.g "probably", "cannot exclude"), which means that the annotations does not necessarily contain noise.

In our training methodology and evaluation, we treated the uncertain labels as negative by mapping them to the negative class, an approach that is commonly used in the related literature. This approach prioritizes specificity and aims to mitigate false overconfident positive predictions. Furthermore, missing labels were also mapped to the negative class.

However, it is important to note that we did not fully discard the uncertainty nature from the workflow. We will see how in later stages of the pipeline the uncertainty is handled through model calibration, stratification of the probability domain into separately divided intervals, and the calculation of clinical likelihood-ratio as an evaluator and a verifier of our stratified zones.

3.3 Target Pathologies

While the base Vision Transformer classifier was trained on all fourteen available labels to leverage the full contextual breadth of the dataset, the explainability and report generation evaluations in this study focus specifically on a subset of five key pathologies: Atelectasis, Cardiomegaly, Consolidation, Edema, and Pleural Effusion. The selection of these specific targets was not arbitrary, but rather driven by three rigorous methodological criteria:

First, alignment with established benchmarks. These five specific pathologies were selected by the organizers of the original CheXpert competition as the primary evaluation metrics. They represent the most critical, actionable, and frequently assessed findings in acute cardiopulmonary care. Restricting the evaluation to this subset ensures that our framework's performance can be contextualized against a broad body of existing literature.

Second, spatial and anatomical diversity. To comprehensively test the robustness of the zone-based explainability framework, it was necessary to select pathologies that exhibit a wide spectrum of visual manifestations. For instance, Cardiomegaly and Pleural Effusion generally present with more predictable, focal boundaries (e.g., lower lung zones and cardiac silhouette), whereas Pulmonary Edema and Atelectasis frequently exhibit diffuse, multi-zone, or patchy patterns. This diversity ensures the dynamic Top-k selection rule is tested under both localized and widespread disease conditions.

Third, label distinctiveness and spatial ambiguity. Evaluating localization mechanisms requires reference findings with relatively distinct radiographic definitions. Several of the remaining nine CheXpert labels were excluded due to high spatial ambiguity or hierarchical overlap. Most notably, "Pneumonia" was excluded because it is fundamentally a clinical diagnosis that manifests radiographically as "Consolidation" or "Lung Opacity." Consequently, bounding box or mask annotations for Pneumonia are highly subjective and overlap extensively with other labels, making it an unreliable target for measuring strict spatial localization accuracy. Similarly, catch-all labels like "Lung Opacity" lack the specificity required to meaningfully evaluate pathology-aware saliency fusion.

3.4 CheXlocalize Dataset for XAI Evaluation

In order to evaluate our explainability framework and our saliency maps, we used the CheXlocalize dataset, which contains localized pathology annotations(polygon coordinates) created by board-certified radiologists, and correspond to the same radiographs from the validation and the test set of the CheXpert dataset[31].

As opposed to bounding-box annotations, the polygons have the advantage of capturing the extent of the diseases more precisely and faithfully. So, in our proposed explainability pipeline, we rasterized these polygons into binary masks. Then, we mapped them onto a grid of 6 partitions by dividing the radiograph vertically(upper/middle/lower zones) and horizontally(left/right zone), and calculating the fraction of pixels that fall into each zone.

What we achieved by dividing the radiograph in six zones was that it allowed us to aggregate the attention of the saliency maps into clinically interpretable and reportable units, which is exactly how radiologists describe the location of an abnormality. So it provided us an efficient way to translate the model's attention into a structured format which is also suitable to be used in an LLM prompt template where the goal is the generation of radiology reports.

Chapter 4

Model Architecture and Training Pipeline

This chapter provides information related to the training pipeline of the multi-label classifier. The specific design allowed for reproducibility because the experiment was configured through an YAML based interface.

4.1 Transformer Model

A Vision Transformer model was employed in this worked and initialized with pretrained weights. The exact configuration is named `vit_large_patch14_clip_336`. The patch size of this ViT is 14 and the input resolution is of 336 X 336 pixels.

To ensure compatibility with the pretrained weights, we replicated the grayscale channel by converting the processing of x-ray images to three-channel format. Also, the classification head was adapted to produce outputs for the labels from the CheXpert dataset.

4.2 Loss Function and Optimization Strategy

Since this work involves a multi-label classification problem, each pathology is modeled as an independent binary prediction.

The loss function selected for training is the Asymmetric Loss (ASL), which is specifically designed to address severe class imbalance in multi-label classification tasks. ASL applies separate focusing parameters to positive and negative samples, reducing the dominance of abundant easy negatives during optimization and improving sensitivity to rare pathological findings.

Missing labels are handled by applying a per-element mask during loss computation, ensuring that unavailable annotations do not contribute to the optimization objective. Uncertain labels are treated according to the uncertainty

policy described in Chapter 3 and are mapped to the negative class prior to loss computation.

The AdamW optimizer was selected for optimization and decoupled weight decay was also used. A cosine annealing learning rate scheduler was also configured with warm-up epochs for early training stabilization.

Furthermore, early stopping was used based on the ROC-AUC metric to prevent overfitting and unnecessary utilization of resources. Then, we saved the best model checkpoint based on the highest mean ROC-AUC performance on all labels, on the validation set.

4.3 Mixup and Exponential Moving Average

A data augmentation technique that was selected to improve the robustness of our model is the Mixup technique. More specifically, a custom implementation was used for our multi-label task, where we linearly interpolate both the radiographs and the label vectors by using samples from a Beta distribution. We disabled the application of this technique in the validation and evaluation step and applied it only in training.

In addition, an Exponential Moving Average (EMA) of model parameters is maintained throughout training. The EMA model is updated after each training epoch and is used for validation and checkpoint selection when enabled. This technique reduces variance in parameter updates and has been shown to improve both stability and final performance, particularly for large-capacity models.

4.4 Automatic Mixed Precision

All experiments are conducted using Automatic Mixed Precision (AMP) to improve computational efficiency and reduce memory consumption. The training pipeline supports both `float16` and `bfloat16` precision, with automatic selection based on hardware capabilities.

AMP is implemented using PyTorch's native autocasting and gradient scaling utilities. Gradient scaling is enabled when using `float16` precision to prevent numerical underflow, while `bfloat16` precision is used when supported by the GPU to avoid the need for scaling. Mixed precision training allows the use of larger batch sizes and accelerates training without sacrificing numerical stability.

4.5 Evaluation Metrics

Model performance is evaluated using threshold-independent metrics suitable for imbalanced multi-label classification. For each pathology, the Receiver Operating Characteristic Area Under the Curve (ROC-AUC) and the Precision–Recall Area Under the Curve (PR-AUC) are computed.

Evaluation is performed on a per-label basis while ignoring missing ground-truth annotations. Aggregate performance is reported as the mean ROC-AUC and mean PR-AUC across all evaluated labels. These metrics are computed during validation at the end of each epoch and are used for model selection and early stopping.

Per-class metrics are additionally computed and stored for detailed analysis, enabling pathology-specific performance inspection and downstream calibration.

4.6 Probability Calibration

After training, the classifier outputs were calibrated using a post-hoc per-class temperature scaling procedure on the held-out validation split. Concretely, for each pathology class c, a separate temperature parameter $T_c > 0$ was learned and applied to the corresponding logit z_c, yielding calibrated probabilities of the form

$$p_c = \sigma\left(\frac{z_c}{T_c}\right),$$

where $\sigma(\cdot)$ denotes the sigmoid function. The temperatures were optimized by minimizing binary cross-entropy on the validation logits, while ignoring missing labels through masking. Calibration quality was assessed before and after scaling using the expected calibration error (ECE) and the Brier score. The goal of this step was not to improve ranking-based discrimination, but to make the predicted probabilities better aligned with empirical risk, so that the threshold-based decision zones introduced in Chapter 5 would have a clearer operational interpretation. These calibrated probabilities are subsequently used for threshold derivation, decision-zone assignment, and constrained report generation.

Chapter 5

Decision-Aware Reporting Framework

5.1 Overview and Motivation

Traditional radiology report generation systems are commonly producing fluent text but using information of the image that was triggered by the model's predictions. Also, post-hoc explainability methods, commonly also known as saliency maps(e.g., gradient or perturbation attributions), rarely are utilized efficiently for report generation as it has been shown that they are not great at localizing abnormalities.

This chapter presents an algorithmic framework that contains a multi-label chest X-ray classifier, the management of the uncertainty of the predicted probabilities of the classifier, and an NLP report generation system that aims to be interpretable and faithful. The core component of the system is to aggregate the saliency outputs into clinically defined and interpretable regions, and to structure the confidence of the language to be used in the radiology report, through a strategy that stratifies the probability domain into separate zones.

5.2 From Image to Zone-Level Evidence

5.2.1 Explainability Methods

The spatial evidence provided by the classifier is essentially generated from the calculation of the saliency maps for each radiograph and pathology. We decided to compute different complementary saliency methods and even fuse them together to test our assumption that each individual method on its own may be unstable or biased toward specific visual features.

We consider attribution techniques from three broad families:

- **Gradient-weighted activation mapping (Grad-CAM++).** Grad-

CAM++ generates class-specific saliency maps by integrating the gradients that backpropagate into a selected convolutional feature layer with that layer's activations. Relative to Grad-CAM, Grad-CAM++ incorporates higher-order weighting factors, enabling it to better capture multiple distinct salient areas and to provide crisper localization, particularly for small or spatially separated abnormalities.

- **Stochastic gradient attribution (GradientShap).** GradientShap estimates Shapley-like feature attributions by averaging gradients computed along randomized interpolation trajectories between a baseline reference and the input image. This typically leads to smoother saliency maps and can lessen sensitivity to minor input perturbations, at the cost of added randomness and increased computational overhead from sampling.

- **Perturbation-based attribution (Occlusion).** Occlusion-based methods evaluate saliency by systematically masking portions of the image and observing the resulting change in the predicted pathology score. These techniques are model-agnostic and more directly connected to causal influence, but they are computationally demanding and sensitive to choices of occlusion patch size and stride, which determine the effective spatial resolution.

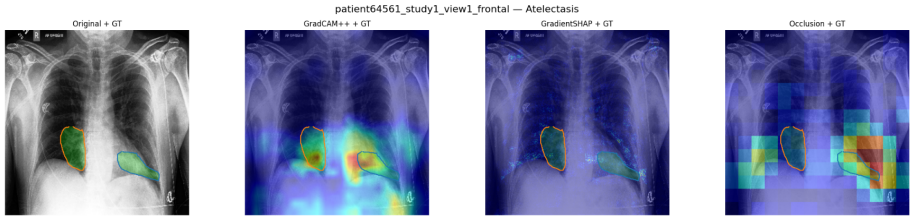


Figure 5.1: Qualitative comparison of saliency methods with CheXlocalize ground-truth overlay for Atelectasis. The original radiograph (left) is shown alongside Grad-CAM++, GradientSHAP, and Occlusion saliency overlays (right). The CheXlocalize polygon annotation is overlaid as a transparent mask with contour to facilitate visual assessment of spatial alignment.

Since saliency maps cannot be directly compared with each other due to the different value ranges and distribution shape differences, we strandarized each map as described below:

- **Spatial resizing.** Saliency maps are resized to a fixed working resolution (e.g., 336×336) to ensure consistency across cases.

- **Normalization.** Each saliency map is normalized to $[0, 1]$.

- **sparsification.** In some experiments, we retained only a fraction of top-p pixels(based on the saliency magnitude), e.g., top-10% or top-20%, and

set the remaining pixels to zero, in order to observe the behavior from keeping only the most salient regions.

5.2.2 Zone-Based Anatomical Representation

Heatmaps from saliency maps are not optimally localizing pathologies in chest radiography, as they are not segmentation models, but our assumption is that they are at least focusing on the clinically correct anatomical regions. So, to structure the spatial evidence of the saliency maps into language that radiologists themselves use to reason, we transform the saliency maps into six non-overlapping regions in order to encode the location where the model is focusing on.

We partition each image into a set of six regions, also called zones, defined by a vertical line that splits the image into two equal sides(left vs right lung) and horizontal lines that divide the lungs into three partitions(upper, middle, lower lungs). So, semantically, the resulted zones are the following:

$$\{\texttt{left upper}, \texttt{left mid}, \texttt{left lower}, \texttt{right upper}, \texttt{right mid}, \texttt{right lower}\}.$$

Zone masks are therefore defined in image coordinates but interpreted in patient-anatomical terms in later reporting. The convention we follow is the same as the standard chest X-ray display, where image-left corresponds to the patient's right side, for example.

Then we implement an Energy-based zone scoring scheme from the saliency maps. Given a normalized saliency map $S \in \mathbb{R}^{H \times W}$ and binary zone masks $\{Z_k\}_{k=1}^{K}$, we compute the saliency "energy" per zone as

$$E_k \;=\; \sum_{(i,j) \in Z_k} S_{ij}.$$

To ensure comparability across images and methods, we convert energies into a normalized distribution

$$p_k \;=\; \frac{E_k}{\sum_{\ell=1}^{K} E_\ell + \epsilon},$$

where ϵ is a small constant to avoid division by zero. The resulting vector $p = (p_k)_{k=1}^{K}$ satisfies $p_k \geq 0$ and $\sum_k p_k \approx 1$ and can be interpreted as a probability-like distribution over zones.

For evaluation, the same six-zone partition is applied to CheXlocalize annotations. For each image and pathology, the radiologist polygon(s) are rasterized into a binary mask $G \in \{0,1\}^{H \times W}$ (treating multiple polygons as their union) and resized to the model resolution. The ground-truth mass in zone k is then computed as:

$$g_k \;=\; \frac{\sum_{(i,j) \in Z_k} G_{ij}}{\sum_{(i,j)} G_{ij} + \epsilon},$$

yielding a normalized distribution $g = (g_k)_{k=1}^{K}$ that summarizes the anatomical distribution of the annotated finding.

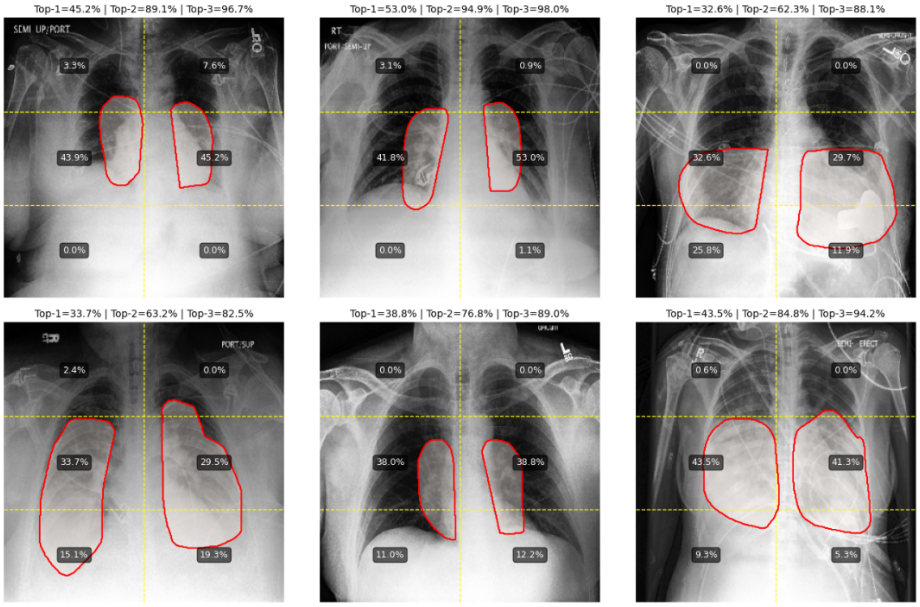


Figure 5.2: CheXlocalize ground-truth zone mass distribution under the six-zone anatomical grid.

From the interpretability and evaluation perspective, this zone-based representation has the following benefits :

- **Clinical readability:** It allow us to express the model's spatial evidence in the same manner that radiologists do in reports(e.g., "right lower lung").
- **Stability:** zone aggregation reduces sensitivity to high-frequency noise by averaging within clinically meaningful regions.
- **Evaluatability:** We can apply the same zone division on the ground truth radiologist annotations, which allows us to compare them against the model's localization evidence

5.2.3 Explainability Fusion and Calibration

Based on the assumption that a single-method saliency may not be sufficient to extract the spatial evidence for the radiology report, we decided to fuse all three saliency methods into a single zone-level evidence distribution.

Let $p^{(m)}(x, c)$ denote the zone distribution produced by method m for image

x and pathology c. Fusion is performed using a weighted average:

$$\tilde{p}(x,c) \;=\; \frac{\sum_{m\in\mathcal{M}(x,c)} w_{c,m}\, p^{(m)}(x,c)}{\sum_{m\in\mathcal{M}(x,c)} w_{c,m}},$$

where $w_{c,m} \geq 0$ is a pathology-specific weight and $\mathcal{M}(x,c)$ is the set of methods available for that case.

The CheXlocalize dataset provides polygon annotations for our target pathologies, which were drawn by certified board radiologists. For each annotations, we rasterized the polygons to create binary masks and resized them to our specified resolution. Then we converted them into a continuous zone-mass distribution using the same zone masks as the model saliency:

$$p^{(\mathrm{GT})}(x,c).$$

This provides a direct supervision signal for localization at the zone level by converting CheXlocalize polygons into a continuous ground-truth zone-mass distribution $p^{(\mathrm{GT})}(x,c)$.

Method weights $\{w_{c,m}\}$ are estimated from a multi-method zone-alignment evaluation on the validation set. For each pathology c and method m, we compute rank-based agreement between the model's zone ordering and the ground-truth distribution, primarily using the normalized top-1 mass capture score

$$\mathrm{NormTop1Capture}(x,c,m) = \frac{\mathrm{Top1Capture}(x,c,m)}{\mathrm{GTTop1Mass}(x,c)}$$

where `Top1Capture` is the GT mass contained in the model's top-ranked zone and `GTTop1Mass` is the GT mass in the dominant GT zone. Weights are then obtained by averaging this score across cases and normalizing across methods per pathology. The given model's name for this fusion method is «Mean Top-1»and a variant of it is named «Mean Top-1$^{\mathbf{2}}$», where the weights are squared to exponentially reward the most reliable method for a specific pathology and suppress noisy methods.

Optionally, we use a composite weighting strategy that additionally incorporates an efficiency term that penalizes overly diffuse evidence by relating approximate top-K GT mass capture to the effective number of GT-positive zones (`gt_n_zones`, clipped to $K \leq 3$). This calibration encourages each pathology to rely more heavily on the saliency method(s) that most consistently rank the clinically relevant zones highest.

We also implemented a strategy that determines dynamically only specific zones that can be used in an LLM prompt because we do not want to overwhelm the model by providing all the zones. Let

$$z^\star = \arg\max_k \tilde{p}_k(x,c), \qquad s^\star = \tilde{p}_{z^\star}(x,c).$$

We select a pathology-specific support set using an α-relative threshold:

$$\mathcal{Z}_\alpha(x,c) \;=\; \{k : \tilde{p}_k(x,c) \geq \alpha\, s^\star\}.$$

Ideally, this rule aims to adapt to the sharpness of the saliency distribution, where focal patterns yields sets containing a few zones, while diffuse patterns yields large sets. It is essentially a mechanism to mitigate hallucination by constraining weaker evidence to not be passed to the language model which will produce the fluent radiology report.

To make the zone evidence passed to the language model both concise and clinically plausible, we additionally calibrate the α-relative zone-selection rule in a pathology-specific manner. Given fused zone scores, a zone is retained if its score is at least α times the maximum fused zone score for that case. Rather than selecting α heuristically, we perform a grid search against CheXlocalize ground truth to identify values that jointly (i) keep the mean number of retained model zones close to the corresponding mean GT zone count and (ii) preserve a minimum GT-mass recall constraint. Figure 5.3 illustrates this calibration behavior for the final Mean Top-1^2 fusion configuration used in the report-generation pipeline. As expected, increasing α yields more compact evidence by reducing the number of retained zones, but also lowers GT-mass recall. The selected pathology-specific values therefore represent a compromise between anatomical sparsity and coverage, ensuring that the structured evidence remains informative without overwhelming the downstream language model.

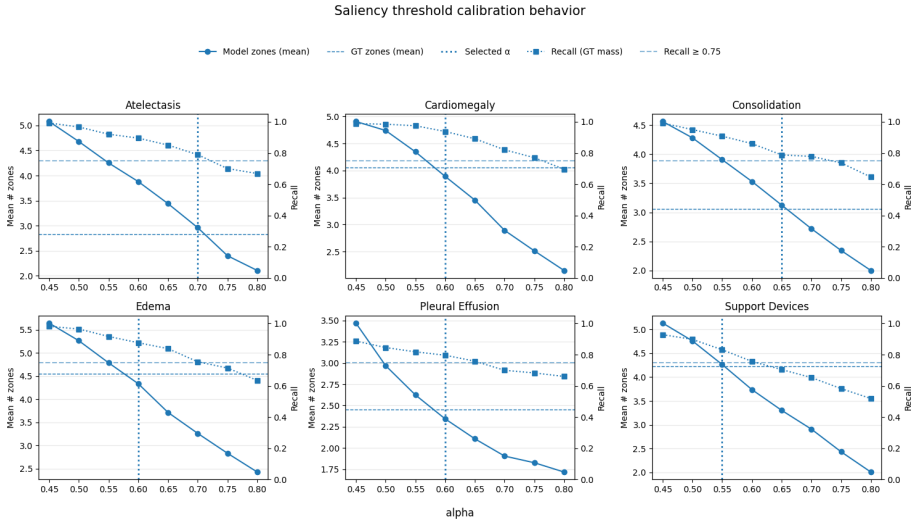


Figure 5.3: Pathology-specific saliency-threshold calibration under the Mean Top-1^2 fusion configuration. For each pathology, increasing α reduces the mean number of retained model zones while also decreasing recall of the ground-truth zone mass. The vertical dotted line marks the selected pathology-specific α, chosen to keep the retained evidence compact while satisfying a minimum recall constraint and remaining close to the corresponding mean ground-truth zone count.

By constraining report generation to zones supported by calibrated multi-method saliency, the framework ties textual claims to an explicit evidence representation.

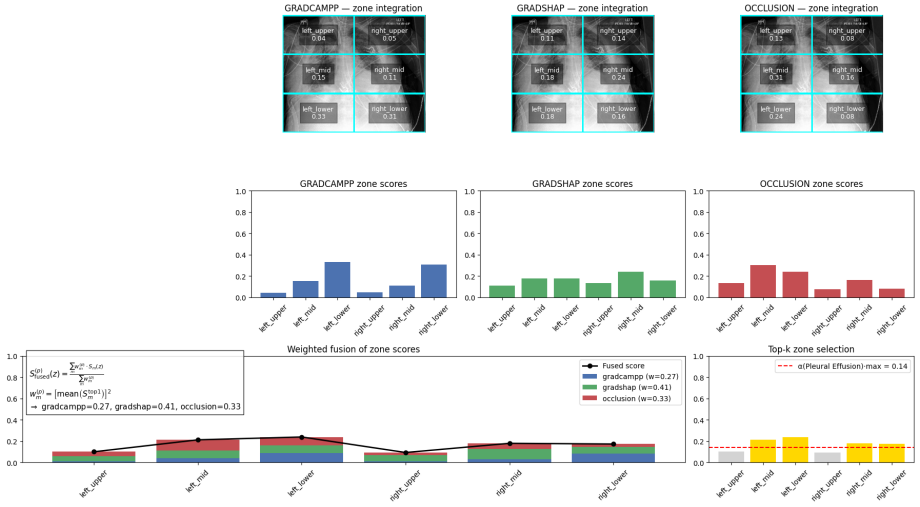


Figure 5.4: Overview of the zone-evidence fusion pipeline used to derive pathology-specific spatial support for report generation. For each saliency method (Grad-CAM++, GradientSHAP, Occlusion), pixel-level attributions are integrated within the six anatomical zones to produce a normalized zone distribution. These per-method zone scores are then combined using pathology-specific weights to obtain a fused zone-evidence distribution. Finally, an α-relative threshold is applied to select the top-supported zones, yielding a compact set of evidence-bearing regions for downstream prompting.

5.3 Decision-Zone Modeling

5.3.1 Motivation

The task of binary classification essentially uses a single operating point to predict the class from the continuous predicted probability. However, radiologists use graded language (e.g., no evidence, cannot exclude, likely, consistent with) to reflect the confidence of their diagnosis when they communicate their observations.

In our attempt to preserve this graded language in our pipeline, we divide the calibrated probability space into separate zones and associate each zone with specific clinical semantics. The bounds for each zone differ for each pathology and are derived from the analysis of sensitivity/specificity behavior on a validation set. The semantic phrasing that corresponds to each zone is described below:

- **Zone 0 (Rule-out / Silent):** predicted probability is below a recall-preserving threshold. So, The model output classifies the observation as negative, and the report remains silent for that pathology
- **Zone 1a (Uncertainty buffer):** predicted probability lies between the rule-out threshold and a balanced operating point, which is the intersection of specificity and sensitivity. In this zone, borderline cases are captured where errors are more common and the model should communicate this uncertainty(e.g., "cannot exclude"), avoiding overconfident positives.
- **Zone 1b (Suggestive):** predicted probability exceeds the balanced operating point but remains below a high-specificity operating point. Cases in this region are more likely true positives than false positives, but still require cautious language (e.g., "suggestive of", "probable").
- **Zone 2 (Rule-in / Conclusive):** predicted probability exceeds a high-specificity operating point, enabling strong language (e.g., "consistent with")

This design aims to mitigate hallucinated certainty on the radiology reports, as the outputs from the same classifier can trigger different diagnostic phrasing based on which zone the predicted probability falls into.

5.3.2 Threshold Derivation

Thresholds are selected separately for each pathology based on a validation set from the train-val split, which contains the ground-truth labels. By partitioning the predicted probability space through the selection of three clinically interpretable thresholds, we aim to construct decision zones that are statistically meaningful. The first threshold aims to preserve high sensitivity (T_{gate}), the second threshold, which is selected at the point where specificity is equal to sensitivity($T_{\text{intersection}}$), aims to capture the ambiguous cases, and finally, the third threshold which is selected at a specified high-specificity point($T_{\text{rule-in}}$), with the purpose of capturing high confident cases.

As far as the uncertain labels in the validation set, we followed the U-zeros policy as we had also done for training the model, where uncertain labels are mapped to the negative class by replacing $-1 \to 0$.

To derive the thresholds, we evaluated a grid of candidate thresholds $t \in [t_{\min}, t_{\max}]$, with 300 evenly spaced steps from 0.01 to 0.99. So, for each t, we defined the classification predictions as $\hat{y} = \mathbb{I}[p \geq t]$ and we calculated the confusion counts (TP, FP, FN, TN), so that we could compute sensitivity and specificity:

$$\text{Sensitivity} = \frac{TP}{TP + FN}, \quad \text{Specificity} = \frac{TN}{TN + FP},$$

The T_{gate} threshold is selected as a rule-out anchor, so it should preserve high recall. So, we first collected all thresholds that satisfy the following recall constraint:

$$\mathcal{T}_{\text{sens}} = \{t \ : \ \text{Recall}(t) \geq \texttt{sens_target}\}.$$

Among these candidates, we specifically identified the minimal recall value satisfies this constraint and then selected the largest threshold achieving that recall value, in cases where there were recall ties.

As far as the uncertainty threshold ($T_{\text{intersection}}$), it is selected as the point where sensitivity and specificity are almost equal at or above T_{gate}:

$$T_{\text{intersection}} = \arg\min_{t \geq T_{\text{gate}}} \left|\text{Recall}(t) - \text{Specificity}(t)\right|.$$

This creates a balanced operating point that tends to lie near the region where false positives and false negatives trade off most evenly, and it defines the upper boundary of the uncertainty buffer (Zone 1a), a zone where the number of negatives are larger than positives.

As far as the high-specificity threshold which aims to report evidence conclusively, it is selected as:

$$T_{\text{rule-in}} = \min\{t \ : \ \text{Specificity}(t) \geq \texttt{spec_target}\}.$$

Where it essentially picks the smallest threshold that allows for the targeted specificity.

So, given these three points, each predicted probability p_i is assigned to a zone:

$$\text{Zone}(p_i) = \begin{cases} \text{Zone 0}, & p_i < T_{\text{gate}}, \\ \text{Zone 1a}, & T_{\text{gate}} \leq p_i < T_{\text{intersection}}, \\ \text{Zone 1b}, & T_{\text{intersection}} \leq p_i < T_{\text{rule-in}}, \\ \text{Zone 2}, & p_i \geq T_{\text{rule-in}}, \end{cases}$$

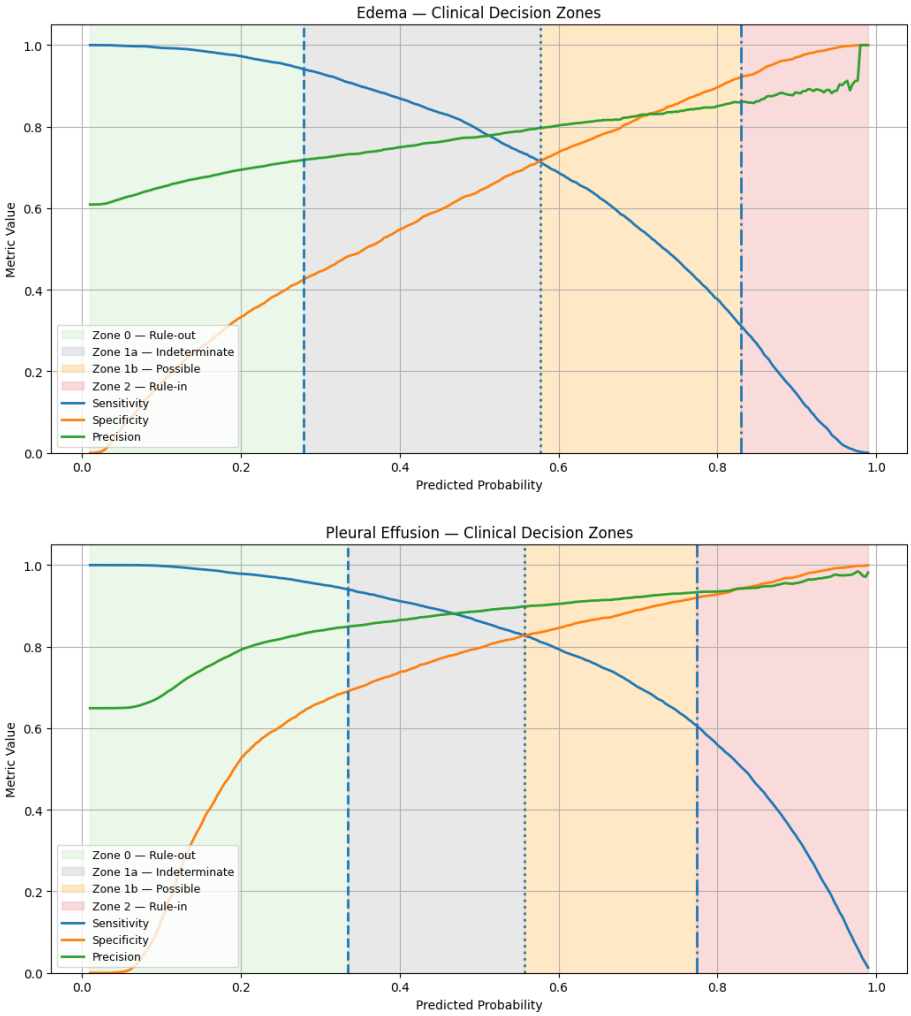


Figure 5.5: Clinical decision zones induced by calibrated probability thresholds on the validation set. Sensitivity, specificity, and precision are plotted as functions of the predicted probability. Vertical markers indicate the derived thresholds T_{gate}, $T_{\text{intersection}}$, and $T_{\text{rule-in}}$, partitioning the probability axis into rule-out (Zone 0), indeterminate (Zone 1a), possible (Zone 1b), and rule-in (Zone 2) regions. Shown for two representative pathologies (Edema and Pleural Effusion).

5.3.3 Likelihood Ratio Estimation in Decision Zones

A metric that is used to quantify the diagnostic strength for the decision zones is the likelihood ratio estimation. It is computed based on the frequency of positives and the negatives that fall into each zone under the under the binary

decision boundary at $T_{\text{intersection}}$.

The actual rule that is used to classify positives and negatives and then to compute TP/FP/FN/TN is:

$$\hat{y}_i = \mathbb{I}[p_i \geq T_{\text{intersection}}].$$

Where, (Zone 1b/2) are interpreted as the positive zones and (Zone 0/1a) are interpreted as the negative zones.

For each sample (y_i, p_i), a zone label is assigned using $(T_{\text{gate}}, T_{\text{intersection}}, T_{\text{rule-in}})$. Then, it increments a per-zone confusion table:

$$\text{zone_counts}[z] \in \{TP, FP, FN, TN\},$$

depending on $(y_i, \hat{y}_i)$.

This results in frequency counts calculation in the zone level and it shows, for example, what fraction of all positives fall into Zone 0, or what fraction of all negatives fall into Zone 2.

Regarding the calculation of the likelihood ratio(LR^-) for the negative zone 0, it is computed as:

$$\text{LR}^-_{Z0} = \frac{P(\text{Zone } 0 \mid D)}{P(\text{Zone } 0 \mid \neg D)} = \frac{FN_{Z0}/P}{TN_{Z0}/N},$$

where $P = \#\{y = 1\}$ and $N = \#\{y = 0\}$ denote the total positives and negatives in the evaluated cohort. Analogously, for Zone 1a it computes an interval likelihood ratio (denoted iLR) using the same form with (FN_{Z1a}, TN_{Z1a}). These statistics characterize how strongly a sample landing in Zone 0 (or Zone 1a) shifts belief toward absence of disease.

Now, as far as the calculation of the Likelihood ratio(LR^+) for the positive zones(Zone 1b and Zone 2), they are computed as:

$$\text{LR}^+_z = \frac{P(z \mid D)}{P(z \mid \neg D)} = \frac{TP_z/P}{FP_z/N}, \qquad z \in \{\text{Zone 1b}, \text{Zone 2}\},$$

5.3.4 Confidence Band Assignment

The primary semantic interpretation is given by the decision zone where the case fall into, but the evaluation results on the likelihood ratios may affect the linguistic phrases, so that the LLM prompt template would later include more reliable evidence.

Additionally, the posterior risk is computed for each zone, which is the posterior probability of the disease given zone membership:

$$P(D \mid z) = \begin{cases} \frac{FN_z}{FN_z + TN_z}, & z \in \{\text{Zone } 0, \text{Zone 1a}\}, \\ \frac{TP_z}{TP_z + FP_z}, & z \in \{\text{Zone 1b}, \text{Zone 2}\}. \end{cases}$$

Finally, we map each pathology-zone pair to a reporting phrase template based on both its decision zone and LR strength. Zone 1b corresponds to intermediate confidence and typically uses cautious language ("suggestive", "probable"), while Zone 2 supports stronger language ("consistent with") when LR^+ is sufficiently high and stable.

Table 5.1: Illustrative confidence mapping using decision zones and likelihood-ratio heuristics.

Decision Zone	**LR statistic used**	**Linguistic confidence**
Zone 0	-	Conclusively negative / silent
Zone 1a	-	Indeterminate / cannot exclude
Zone 1b	LR^+	Suggestive / probable
Zone 2	LR^+	Conclusive / consistent with

To summarize, the construction of those four zones maps the calibrated probabilities of the classifier to an appropriate use of language for reporting. Thresholds are derived per pathology through sensitivity and specificity analysis and likelihood ratios estimates validate and characterize the diagnostic meaning of the positive zones so that the phrasing is clinically aligned.

5.3.5 Constrained Radiology Report Generation

The final step of the framework is to generate a radiology report by utilizing the calibrated classifier outputs and the XAI evidence that is provided at the zone level. The main goal is to harness an Large Language Model(LLM) that produces text that is constrained based on the evidence it is provided only so its behavior is consistent. In order to make this possible, we construct an appropriate prompt template and the selection of our LLM is the gpt-4o-mini model.

For each image and pathology, the pipeline creates a structured record that contains the following:

- **Calibrated probability** p (model score after calibration),
- **Decision zone** $z \in \{\text{Zone 0}, \text{Zone 1a}, \text{Zone 1b}, \text{Zone 2}\}$ derived from pathology-specific thresholds,
- **Localization evidence** in the form of a small Top-K set of coarse anatomical zones (derived from fused saliency).

The calibrated probability is first associated to a decision zone based on the pathology-specific thresholds we previously selected. If it is associated to zone 0, then the pathology will not be mentioned in the prompt and it is excluded from the prompt evidence.

A common failure characteristic in medical text generation is the falsely overconfident or hallucinated statements when the classifier outputs are not well

calibrated. To mitigate this, likelihood ratios are used to modulate the linguistic confidence within a decision zone. As far as the selection of the vocabulary used for the confidence level, the generation pipeline restricts the LLM to a standardized set of confidence levels:

{Conclusively Negative, Indeterminate, Suggestive, Probable, Conclusive}.

Zone 0 always maps to *Conclusively Negative* and Zone 1a always maps to *Indeterminate*, preventing accidental escalation of language for borderline probabilities.

For Zone 1b and Zone 2, the system uses likelihood ratio thresholds inspired by the Jaeschke-style evidence strength heuristic (e.g., larger LR^+ implies stronger diagnostic support)[32]. In practice, if the raw LR estimate is unstable (e.g., infinite due to $FP = 0$, missing, or derived from small counts), the policy falls back to a more conservative choice (e.g., a lower bound if available, or downgrading confidence).

In the cases where the pathology is reportable(Zone 1a/1b/2), the prompt includes the Top-K list of anatomical zones(e.g., two zones for atelectasis and consolidation, three for pleural effusion and edema) based on the rule we previously described. Essentially, the top-K zones correspond to the regions where the model's attention were the highest.

Finally, the LLM prompt includes the following information:

- **Patient metadata** (age, sex, projection),
- **Structured evidence JSON** listing only reportable pathologies and their permitted confidence bands,
- **A confidence-band policy** with allowable phrase categories per confidence level.

LLM prompt template used for constrained radiology report generation (example case)

```
You are an expert thoracic radiologist. Write FINDINGS and IMPRESSION
sections for a single chest X-ray.

PATIENT
- Age: 84
- Sex: Female
- Projection: AP

You must rely ONLY on the following structured evidence (POSITIVE FINDINGS ONLY):

{
  "Atelectasis": {
    "confidence_level": "Indeterminate",
    "top_zones": [
      "left lower lung",
      "right lower lung"
    ]
  },
  "Cardiomegaly": {
    "confidence_level": "Suggestive"
  },
  "Consolidation": {
    "confidence_level": "Suggestive",
    "top_zones": [
      "left lower lung",
      "left mid lung"
    ]
  },
  "Edema": {
    "confidence_level": "Conclusive",
    "top_zones": [
      "right mid lung",
      "right lower lung",
      "left mid lung"
    ]
  },
  "Pleural Effusion": {
    "confidence_level": "Conclusive",
    "top_zones": [
      "left lower lung",
      "right lower lung",
      "left mid lung"
    ]
  }
}
IMPORTANT INTERPRETATION NOTE:
- Any listed top_zones represent regions of highest model attention.
- They are NOT exact boundaries, ranked certainty, or mutually exclusive.
- Multiple zones may reflect diffuse, bilateral, or multi-regional involvement.
RULES:
1) Use standard radiology language.
2) FINDINGS: include ONLY evidence-supported abnormalities.
3) IMPRESSION: 13 concise sentences summarizing ONLY abnormalities mentioned in
     FINDINGS.
4) Each pathology includes a confidence_level field indicating the ONLY
   certainty band you may use for that pathology.

Confidence bands:
- Indeterminate: (e.g., "equivocal", "cannot exclude", "uncertain finding")
- Suggestive: (e.g., "possible", "could represent", "suspicious for")
- Probable: (e.g., "likely", "consistent with", "probable")
```

```
- Conclusive: State the finding as a definitive fact (e.g., "There is...", "Seen
     is...").

IMPORTANT:
- If confidence_level is "Indeterminate":
  - You may ONLY use uncertainty phrases
  - Do NOT imply presence or likelihood

You MUST:
- match the language strength to confidence_level
- NOT use language stronger than the confidence_level permits


5) Do NOT introduce diseases outside the target conditions.
6) Do NOT give recommendations.

OUTPUT FORMAT:
FINDINGS:
...

IMPRESSION:
...
```

Figure 5.6: Example of the constrained prompt template provided to the LLM. The prompt includes structured pathology evidence gated by decision zones and top-k saliency-supported regions, and strict negative/forbidden rules to reduce unsupported findings in the generated report.

To summarize, the constrained radiology report generation works by (i) using likelihood ratios to modulate confidence phrasing, (ii) selecting only the top-k anatomical zones that the model is most focused on, (iii) and applying a post-processing step on the LLM outputs to further reduce hallucinations that may occur. All these mechanisms aim to generate radiology reports that are evidence-grounded to face the black-box nature of neural networks.

Chapter 6

Evaluation

6.1 Evaluation Setup

This chapter evaluates the explainable decision-aware reporting pipeline we have proposed. First, we evaluate the ability of the explainable signals, either individually, or fused, to localize the abnormalities in clinically relevant regions by using the given radiologist-annotated ground truth. Then, we assess whether the generated radiology reports are faithful to the structured evidence.

6.1.1 Datasets

Three data sources are used for evaluation. CheXpert-small is used to train the multi-label chest X-ray classifier and to compute pathology probabilities which they are been calibrated then. CheXlocalize provides the radiologist annotated polygons that are used for spatial evaluation. We only consider frontal views(AP / PA) to align with the lung-zone anatomical partition, so the lateral radiographs are excluded. Also, CheXpert-Plus is used to evaluate the generated text of the LLM against the referenced text. Table 6.1 summarizes the role of each dataset and the resulting evaluation sample sizes.

Table 6.1: Datasets used for evaluation and their roles in the pipeline.

Dataset	**Role**	**Ground truth type**	**Size (N)**
CheXpert-small	training	image-level labels	224,316
CheXlocalize (val)	localization eval	pathology polygons	171 (frontal)
CheXpert Plus (val)	report eval	report text / derived labels	200

CheXlocalize provides pathology annotations for 171 frontal views out of the 202 frontal radiographs that exist in the radiologist validation split(lateral views excluded), which correspond to 279 annotated image–pathology instances across the evaluated findings.

Paired radiology reports are available only for the validation reference subset used in this thesis; the held-out test split used for image-level evaluation does not provide corresponding report text in our setup, and therefore report-level evaluation is restricted to the validation set.

6.1.2 Ground Truth Definitions

Evaluation is performed on five clinically important findings: Atelectasis, Cardiomegaly, Consolidation, Edema, Pleural Effusion. For classification, ground truth corresponds to CheXpert image-level labels (with uncertain labels handled as described in Chapter 4). For localization, ground truth consists of CheXlocalize polygons which are rasterized into binary masks, resized to the model input resolution, and summarized into a continuous six-zone distribution (left/right $\times$ upper/mid/lower). For report-level evaluation, reference information is obtained from the associated report text and/or labelers applied to the reference reports.

6.1.3 Implementation notes

All evaluations are performed using the same image preprocessing pipeline as used during inference (input resolution: [e.g., 336×336]), ensuring consistent alignment between model evidence and spatial ground truth. CheXlocalize polygons are rasterized and resized using the same geometric transforms as applied to the input images to avoid systematic misalignment. Decision-zone thresholds are derived from calibrated probabilities on a held-out validation split (Chapter 5), and the same fixed thresholds are used throughout this evaluation to avoid information leakage.

6.2 Classifier performance

Table 6.2: Discrimination performance on the validation set for the five target pathologies used throughout the thesis.

Pathology	**ROC-AUC**	**PR-AUC**
Atelectasis	0.788	0.599
Cardiomegaly	0.788	0.696
Consolidation	0.906	0.558
Edema	0.921	0.713
Pleural Effusion	0.917	0.806

Table 6.3: Discrimination performance on the held-out test set for the five target pathologies used throughout the thesis.

Pathology	**ROC-AUC**	**PR-AUC**
Atelectasis	0.823	0.601
Cardiomegaly	0.890	0.776
Consolidation	0.875	0.310
Edema	0.893	0.524
Pleural Effusion	0.951	0.840

These results indicate generally strong discrimination for the target findings, supporting the use of calibrated probabilities for decision-zone gating and downstream evidence construction in Chapters 5–6. PR-AUC varies more across splits due to its stronger dependence on class prevalence and imbalance.

6.3 Localization Evaluation

Localization is evaluated on the radiologist validation subset with polygon annotations. For each image–pathology instance with available ground truth, the annotated polygons are rasterized into a binary mask, resized to match the model input resolution, and summarized into a six-zone anatomical grid (left-/right × upper/mid/lower). Explainability maps are produced per pathology and compared under three single-method baselines (Grad-CAM++, GradSHAP, Occlusion) and several fusion variants. Fusion strategies include a simple baseline fusion as well as score-composition heuristics (Composite, Mean Top-1, Mean Top-1^2), optionally combined with saliency sparsification (retaining the top 10%, 20%, or 30% most salient pixels prior to zone aggregation). As far as the naming of the fusion experiments, we note that Baseline Fusion denotes a simple weighted aggregation of zone-level scores across saliency methods, composite denotes a heuristic fusion rule that combines method signals using pathology-dependent reliability weights, Mean Top-1 denotes pathology-specific fusion where method weights are proportional to each method's mean normalized Top-1 alignment with CheXlocalize (rank-based), and the Mean Top-1^2 variant which squares these weights to emphasize the strongest method per pathology.

6.3.1 Localization Metrics

We report complementary metrics that capture how well the method concentrates attention on clinically relevant zones and how accurately it recovers the ground-truth dominant region.

GT abnormality retrieval at Top-k. Let $\mathbf{g} \in \mathbb{R}^6$ denote the ground-truth zone distribution obtained from the polygon mask (zone-wise mass), and let

$\mathbf{s} \in \mathbb{R}^6$ denote the predicted zone scores derived from a saliency map. Let π_k be the indices of the k highest-scoring predicted zones. The Top-k GT mass capture is defined as

$$\text{Mass@Top-k} = \sum_{z \in \pi_k} g_z, \tag{6.1}$$

which measures the fraction of ground-truth annotated area retrieved by the top k ranked zones.

Relative GT abnormality retrieval. Since the total mass is most usually located on more than one lung zones and the ground truth usually has diffuse patterns, we decided to also report a normalized variant of Mass@Top-k to improve comparability. Two separate settings are calculated. In one setting, B_k corresponds to the mean fraction of GT mass contained in the top-k GT-ranked zones averaged across the evaluated pathologies (computed from the GT rank-1/2/3 fractions). The other setting is focused on the per-pathology level, where B_k is computed analogously using the rank-wise GT statistics for the corresponding pathology.

The normalized score is then

$$\text{NormMass@Top-k} = \frac{\text{Mass@Top-k}}{B_k}, \tag{6.2}$$

yielding values in $[0, 1]$, where 1.0 indicates performance matching the reference Top-k GT bound under the corresponding setting.

The top-1 zone essentially corresponds to the region where the largest fraction of the annotated area is located at. Figure 6.1 demonstrates the mean ground-truth zone coverage fraction for each pathology and it is observed that pleural effusion is highly concentrated in the dominant zone with an average coverage of 72% while the rest of the pathologies are more diffuse and their average top-1 coverage lies in the range of approximately 40 - 50 %. These ground-truth zone statistics motivate reporting results at Top-1/Top-2/Top-3, justify why normalizing Mass@Top-k by rank-wise GT statistics makes the comparison more fair, and suggests that the extraction of the XAI evidence should include more than one anatomical lung zone to be reported in a generated radiology report.

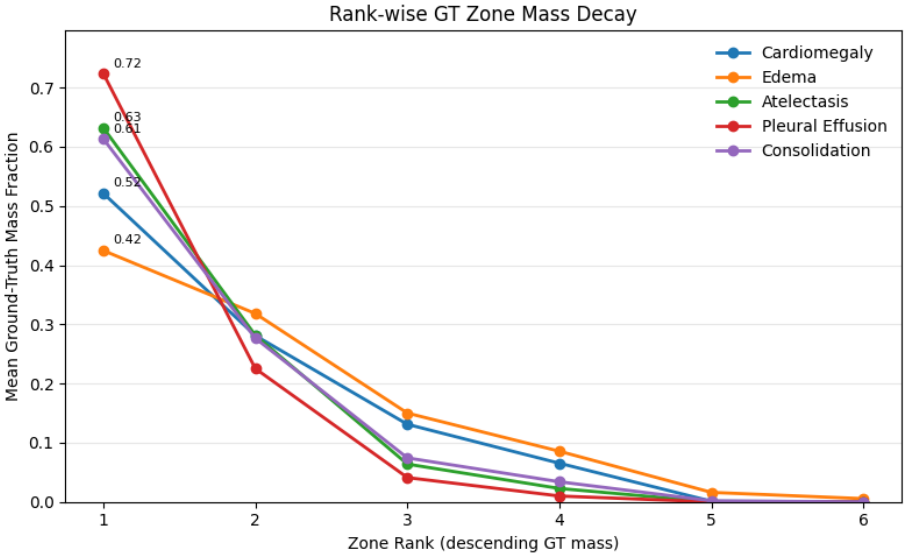


Figure 6.1: Rank-wise decay of the mean ground-truth zone coverage fraction (CheXlocalize validation). For each pathology, zones are sorted by decreasing ground-truth coverage and averaged across cases.

Top-k zone accuracy. To assess whether the method identifies the dominant region, we compute whether the ground-truth dominant zone (argmax of $\mathbf{g}$) lies within the predicted Top-k zones (from $\mathbf{s}$). This is reported both over all instances and over the clinically unilateral subset.

Dominant laterality accuracy. We additionally evaluate whether the explanation is consistent with the ground-truth side of involvement. For each instance, we infer the model's dominant laterality from the Top-k predicted zones by summing predicted scores over left-lung zones and right-lung zones and selecting the larger side. For ground truth, we report two settings: (i) a clinically unilateral subset where laterality is defined at the instance level (left vs right) and bilateral cases are excluded; and (ii) an all-cases setting where ground-truth laterality is defined by the dominant ground-truth mass across left vs right zones. Dominant laterality accuracy is the fraction of cases where the model-predicted dominant side matches the corresponding ground-truth laterality definition. Dominant laterality accuracy is reported only for pathologies with a clinically meaningful unilateral/bilateral interpretation under the lung-zone grid. In our evaluation, this includes Atelectasis, Consolidation, and Pleural Effusion. Edema is excluded in that metric since it is predominantly a bilateral pathology(Table 6.6).

6.3.2 Results: Overall Localization Performance

Table 6.11 summarizes GT mass capture at Top-k across all annotated instances ($N = 279$), while Table 6.5 reports the normalized variant.

Across methods, GradSHAP substantially outperforms Grad-CAM++ and Occlusion, achieving a normalized mass capture of 0.653 (Top-1), 0.707 (Top-2), and 0.813 (Top-3), compared to 0.470/0.539/0.750 for Grad-CAM++ and 0.525/0.576/0.691 for Occlusion. Fusion methods consistently improves over single methods. The baseline fusion, that is, the resulting method from the weighted average combination of the three saliency techniques with equal weights, achieves 0.651/0.723/0.827 (normalized Top-1/2/3). Among fusion heuristics, Mean Top-1^2 yields the strongest overall normalized mass capture, reaching 0.662 (Top-1), 0.757 (Top-2), and 0.859 (Top-3), indicating that an elitist fusion that upweights the strongest per-pathology method signals (via squared Top-1–based weights) yields the most faithful spatial grounding, while still supporting multi-zone retrieval when evaluated at Top-2/Top-3. Here, per-pathology method weights are derived from rank-based CheXlocalize alignment (mean normalized Top-1), with squaring (power=2) used to boost the best-performing method per pathology.

Sparsifying the saliency map (top-10%, top-20%, top-30%) exhibits a mixed effect. While sparsification can slightly improve certain configurations at higher k (e.g., baseline fusion Top-3), it can also reduce Top-1 performance, suggesting a trade-off between removing noisy activations and preserving weak but consistent evidence for diffuse findings.

Table 6.4: Ground-truth pathology coverage captured by the model's top-K predicted lung zones.

Experiment	Top-1	Top-2	Top-3
GradSHAP	0.380900	0.607900	0.773500
GradSHAP top-10%	0.366400	0.615500	0.766800
GradSHAP top-20%	0.369700	0.603300	0.767900
GradSHAP top-30%	0.373100	0.607000	0.770000
GradCAM++	0.274200	0.463200	0.713400
Occlusion	0.305800	0.495200	0.657000
Occlusion top-10%	0.319100	0.524700	0.674800
Occlusion top-20%	0.315000	0.518300	0.680200
Occlusion top-30%	0.315200	0.512500	0.688400
Baseline Fusion	0.379500	0.621000	0.786600
Baseline Fusion top-10%	0.364000	0.608700	0.791900
Baseline Fusion top-20%	0.375900	0.626600	0.797800
Baseline Fusion top-30%	0.376300	0.629900	0.801900
Composite	0.381600	0.628900	0.809100
Mean Top-1	0.384400	<u>0.641900</u>	<u>0.810200</u>
Mean Top-1²	**0.386200**	**0.650200**	**0.817100**
Mean Top-1² top-10%	0.384700	0.631400	0.793600
Mean Top-1² top-20%	<u>0.383800</u>	0.634300	0.800300

Table 6.5: Ground-truth coverage normalized by the maximum achievable GT mass at each Top-K level.

Experiment	Top-1	Top-2	Top-3
GradSHAP	0.653400	0.707300	0.812900
GradSHAP top-10%	0.628500	0.716100	0.805800
GradSHAP top-20%	0.634100	0.701900	0.807000
GradSHAP top-30%	0.640000	0.706300	0.809200
GradCAM++	0.470300	0.538900	0.749700
Occlusion	0.524500	0.576200	0.690500
Occlusion top-10%	0.547400	0.610500	0.709200
Occlusion top-20%	0.540300	0.603000	0.714800
Occlusion top-30%	0.540700	0.596300	0.723400
Baseline Fusion	0.651000	0.722500	0.826600
Baseline Fusion top-10%	0.624400	0.708200	0.832200
Baseline Fusion top-20%	0.644800	0.729100	0.838400
Baseline Fusion top-30%	0.645500	0.732900	0.842700
Composite	0.654600	0.731700	0.850300
Mean Top-1	0.659400	0.746900	0.851500
Mean Top-1^2	**0.662400**	**0.756500**	**0.858700**
Mean Top-1^2 top-10%	0.659900	0.734600	0.834000
Mean Top-1^2 top-20%	0.658300	0.738000	0.841000

6.3.3 Results: Zone Accuracy and Laterality

Zone-retrieval accuracy results are shown in Table 6.7. Consistent with the mass-capture results, fusion variants outperform single-method baselines. Mean Top-1^2 achieves the best overall Top-k zone accuracy, with 0.480 (Top-1), 0.756 (Top-2), and 0.867 (Top-3). A Top-1 zone accuracy of 48.0 means that in approximately 48% of cases, the model's single highest scoring zone aligns with the ground-truth dominant zone (the zone with the highest GT coverage). Allowing the model to return the two most salient zones substantially increases the chance of retrieving the dominant GT zone to a 75.6% accuracy, whereas using Top-3 model's predicted further raises it to 86.7%, indicating that for most instances the ground-truth dominant region is recovered when the explanation is permitted to represent a small set of zones rather than a single location. This pattern is clinically plausible for findings that are multi-regional or bilaterally distributed, and it supports the use of multi-zone evidence (Top-2/Top-3) in the downstream report generator instead of committing to a single zone.

On the clinically unilateral subset ($N = 113$) for Atelectasis, Consolidation, and Pleural Effusion, where left-right localization is easier to interpret due to the lower prevalence of bilateral cases (as shown in Table 6.6), fusion improves dominant laterality accuracy relative to single-method baselines(Table 6.9). Dominant laterality accuracy is highest for Mean Top-1^2 at 0.779, followed by Mean Top-1 (0.752) and fusion variants such as Baseline Fusion top-10% (0.743), indicating that pathology-aware fusion yields more side-consistent evidence than any individual saliency method.

Table 6.6: Distribution of bilateral cases per pathology in the CheXlocalize validation set.

pathology	N_{total}	$N_{bilateral}$	Bilateral (%)
Edema	42	39	92.9
Atelectasis	75	30	40.0
Consolidation	32	11	34.4
Pleural Effusion	64	17	26.6

To assess laterality consistency more broadly, we also report dominant laterality accuracy over both unilateral and bilateral cases. The same ordering is preserved: Mean Top-1^2 remains best (0.749), followed by Mean Top-1^2 top-10% (0.737) and Mean Top-1 (0.731). Overall, these results suggest that fusion improves laterality grounding both in strictly unilateral cases and when laterality is defined by the dominant GT mass.

Also, in Table 6.8, we show the results from the top-k zone accuracy specifically on unilateral cases only, and we notice that the results do not differ significantly compared to the results from all cases. For example, the top-1 zone accuracy from the best experiment (Mean Top-1^2) is 52.21% compared to 48% on all cases.

Table 6.7: Top-K zone accuracy across localization strategies.

Experiment	**Top-1**	**Top-2**	**Top-3**
GradSHAP	0.4767	0.6918	0.8423
GradSHAP top-10%	0.4516	0.7133	0.8387
GradSHAP top-20%	0.4588	0.6846	0.8423
GradSHAP top-30%	0.4659	0.6882	0.8387
GradCAM++	0.2975	0.4480	0.7312
Occlusion	0.3907	0.5591	0.7276
Occlusion top-10%	0.4014	0.5986	0.7276
Occlusion top-20%	0.3907	0.5842	0.7348
Occlusion top-30%	0.3943	0.5842	0.7527
Baseline Fusion	0.4444	0.7061	0.8315
Baseline Fusion top-10%	0.4409	0.6774	0.8495
Baseline Fusion top-20%	0.4373	0.6989	0.8459
Baseline Fusion top-30%	0.4373	0.7133	0.8387
Composite	0.4624	0.7276	0.8602
Mean Top-1	0.4624	0.7384	0.8638
Mean Top-$\mathbf{1^2}$	**0.4803**	**0.7563**	**0.8674**
Mean Top-$\mathbf{1^2}$ top-10%	0.4695	0.7276	0.8566
Mean Top-$\mathbf{1^2}$ top-20%	0.4695	0.7384	0.8566

Table 6.8: Top-k zone accuracy on clinically unilateral cases ($N = 113$).

Experiment	**Top-1**	**Top-2**	**Top-3**
GradSHAP	0.5044	0.6637	0.7965
GradSHAP top-10%	0.4867	0.6726	0.7699
GradSHAP top-20%	0.4867	0.6549	0.7699
GradSHAP top-30%	0.4956	0.6637	0.7876
GradCAM++	0.3186	0.5487	0.7699
Occlusion	0.4071	0.5752	0.7434
Occlusion top-10%	0.4071	0.6106	0.7257
Occlusion top-20%	0.4159	0.6283	0.7522
Occlusion top-30%	0.4159	0.6195	0.7611
Baseline Fusion	0.5133	0.7434	0.8230
Baseline Fusion top-10%	0.4779	0.6726	0.8496
Baseline Fusion top-20%	0.4956	0.7168	0.8319
Baseline Fusion top-30%	0.5044	0.7345	**0.8584**
Composite	0.5044	0.7168	0.8407
Mean Top-1	0.5133	0.7345	0.8319
Mean Top-1^2	**0.5221**	**0.7522**	0.8407
Mean Top-1^2 top-10%	**0.5221**	0.6903	0.8142
Mean Top-1^2 top-20%	**0.5221**	0.7080	0.8319

Table 6.9: Dominant laterality accuracy for unilateral and bilateral cases(pathologies : Atelectasis, Consolidation, Pleural Effusion)

Experiment	Acc (unilateral, GT_{inst})	Acc (all, GT_{mass})
GradSHAP	0.6903	0.6725
GradSHAP top-10%	0.6637	0.6433
GradSHAP top-20%	0.6726	0.6550
GradSHAP top-30%	0.6814	0.6608
GradCAM++	0.7080	0.7018
Occlusion	0.6372	0.6433
Occlusion top-10%	0.6460	0.6491
Occlusion top-20%	0.6549	0.6608
Occlusion top-30%	0.6283	0.6433
Baseline Fusion	0.7168	0.7076
Baseline Fusion top-10%	0.7434	0.7251
Baseline Fusion top-20%	0.7168	0.7076
Baseline Fusion top-30%	0.7345	0.7193
Composite	0.7257	0.7193
Mean Top-1	0.7522	0.7310
Mean Top-1^2	**0.7788**	**0.7485**
Mean Top-1^2 top-10%	<u>0.7611</u>	<u>0.7368</u>
Mean Top-1^2 top-20%	0.7434	0.7251

6.3.4 Per-Pathology Behavior

Per-pathology normalized mass capture (Table 6.10) highlights clinically plausible heterogeneity. Cardiomegaly exhibits consistently high normalized capture across strong methods (often exceeding ≈ 0.67 at Top-1 and ≈ 0.80 at Top-2), reflecting a relatively coherent anatomical target. In contrast, diffuse or multi-regional patterns (e.g., Edema) show larger sensitivity to the fusion rule and sparsification level, consistent with broader spatial extent.

The best-performing strategy varies by pathology: for Atelectasis and Consolidation, Mean Top-1^2 provides the strongest Top-2/Top-3 capture, whereas for Pleural Effusion, Mean Top-1 is strongest at Top-1/Top-2 and baseline fusion top-30% is strongest at Top-3. This pathology dependence is expected because the fusion weights are derived per pathology from rank-based alignment statistics (mean normalized Top-1), with optional "elitism" from squaring to prioritize the strongest method.

Finally, performance for Consolidation ($N = 32$) should be interpreted more cautiously due to the smaller sample size, and sparsification effects are not uniform across findings (e.g., improving Top-1 for Edema under some settings while degrading Top-1 for Atelectasis), suggesting that noise suppression trades off against preserving weak distributed evidence.

Table 6.10: Per-pathology normalized ground-truth mass capture at Top-K. Best-performing method per pathology and Top-K is shown in bold, with the second-best underlined. Values are normalized by the reference Top-K GT bound derived from rank-wise GT statistics.

Pathology	**Experiment**	**Top-1**	**Top-2**	**Top-3**	**N cases**
Atelectasis	Baseline Fusion	<u>0.6026</u>	0.6301	0.7585	75
Atelectasis	Baseline Fusion top-10%	0.5025	0.5885	0.7445	75
Atelectasis	Baseline Fusion top-20%	0.5672	0.6315	0.7363	75
Atelectasis	Baseline Fusion top-30%	0.5805	0.6495	0.7670	75
Atelectasis	Composite	0.5883	0.6037	<u>0.7858</u>	75
Atelectasis	GradCAM++	0.4757	0.5217	0.7348	75
Atelectasis	GradSHAP	**0.6113**	0.6394	0.7792	75
Atelectasis	GradSHAP top-10%	0.5561	0.6252	0.7494	75
Atelectasis	GradSHAP top-20%	0.5633	0.6303	0.7527	75
Atelectasis	GradSHAP top-30%	0.5783	0.6445	0.7741	75
Atelectasis	Mean Top-1	0.5825	<u>0.6528</u>	0.7831	75
Atelectasis	Mean Top-1^2	0.5897	**0.6785**	**0.8107**	75
Atelectasis	Mean Top-1^2 top-10%	0.5989	0.6193	0.7488	75
Atelectasis	Mean Top-1^2 top-20%	0.5825	0.6283	0.7578	75
Atelectasis	Occlusion	0.3470	0.4200	0.5730	75
Atelectasis	Occlusion top-10%	0.3690	0.4705	0.6210	75
Atelectasis	Occlusion top-20%	0.3698	0.4537	0.6249	75
Atelectasis	Occlusion top-30%	0.3489	0.4330	0.6309	75
Cardiomegaly	Baseline Fusion	0.6783	0.7995	0.8864	66

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Pathology	Experiment	Top-1	Top-2	Top-3	N cases
Cardiomegaly	Baseline Fusion top-10%	0.6604	0.7969	0.8801	66
Cardiomegaly	Baseline Fusion top-20%	0.6746	0.7983	0.8976	66
Cardiomegaly	Baseline Fusion top-30%	0.6637	0.8194	0.8853	66
Cardiomegaly	Composite	0.6782	0.8129	0.8950	66
Cardiomegaly	GradCAM++	0.4998	0.5359	0.7844	66
Cardiomegaly	GradSHAP	0.6804	0.8152	0.8788	66
Cardiomegaly	GradSHAP top-10%	0.6593	0.7956	0.8606	66
Cardiomegaly	GradSHAP top-20%	0.6774	0.7970	0.8791	66
Cardiomegaly	GradSHAP top-30%	0.6745	0.8047	0.8768	66
Cardiomegaly	Mean Top-1	0.6708	0.8134	0.8982	66
Cardiomegaly	Mean Top-1^2	0.6794	0.8298	0.8921	66
Cardiomegaly	Mean Top-1^2 top-10%	0.6839	0.8297	**0.8990**	66
Cardiomegaly	Mean Top-1^2 top-20%	**0.6912**	**0.8439**	0.8945	66
Cardiomegaly	Occlusion	0.5704	0.5587	0.6588	66
Cardiomegaly	Occlusion top-10%	0.6117	0.6397	0.6886	66
Cardiomegaly	Occlusion top-20%	0.5698	0.6035	0.6881	66
Cardiomegaly	Occlusion top-30%	0.5756	0.6056	0.6845	66
Consolidation	Baseline Fusion	0.5692	0.7546	0.8587	32
Consolidation	Baseline Fusion top-10%	0.6190	0.7419	0.8882	32
Consolidation	Baseline Fusion top-20%	0.6188	0.7473	0.8798	32
Consolidation	Baseline Fusion top-30%	0.6188	0.7468	0.8882	32
Consolidation	Composite	0.5948	0.7645	0.8794	32
Consolidation	GradCAM++	0.4018	0.5201	0.8056	32
Consolidation	GradSHAP	0.6260	0.6931	0.7935	32
Consolidation	GradSHAP top-10%	0.6402	0.7135	0.7963	32
Consolidation	GradSHAP top-20%	0.6402	0.7004	0.7935	32
Consolidation	GradSHAP top-30%	0.6291	0.7004	0.7935	32
Consolidation	Mean Top-1	0.5948	0.7619	0.8856	32
Consolidation	Mean Top-1^2	0.6459	**0.7907**	**0.9065**	32
Consolidation	Mean Top-1^2 top-10%	0.6375	0.7618	0.8346	32
Consolidation	Mean Top-1^2 top-20%	0.6356	0.7604	0.8625	32
Consolidation	Occlusion	0.5901	0.6517	0.7253	32
Consolidation	Occlusion top-10%	0.5982	0.6447	0.7265	32
Consolidation	Occlusion top-20%	0.5967	0.6633	0.7816	32
Consolidation	Occlusion top-30%	**0.6691**	0.6763	0.7739	32
Edema	Baseline Fusion	0.6309	0.6386	0.7636	42
Edema	Baseline Fusion top-10%	0.7231	0.6697	0.8154	42
Edema	Baseline Fusion top-20%	0.6877	0.7011	0.8138	42
Edema	Baseline Fusion top-30%	0.6341	0.6676	0.7903	42
Edema	Composite	0.7141	0.7309	**0.8574**	42
Edema	GradCAM++	0.2886	0.3002	0.5760	42
Edema	GradSHAP	0.7238	0.7044	0.7854	42
Edema	GradSHAP top-10%	0.6976	**0.7652**	0.8097	42
Edema	GradSHAP top-20%	0.6962	0.7193	0.7956	42
Edema	GradSHAP top-30%	0.6962	0.7069	0.7837	42
Edema	Mean Top-1	0.7315	0.7376	0.8529	42

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Pathology	Experiment	Top-1	Top-2	Top-3	N cases
Edema	Mean Top-1^2	0.7231	0.7250	0.8380	42
Edema	Mean Top-1^2 top-10%	0.7211	<u>0.7448</u>	0.8423	42
Edema	Mean Top-1^2 top-20%	**0.7351**	0.7251	0.8100	42
Edema	Occlusion	0.6244	0.6387	0.7021	42
Edema	Occlusion top-10%	0.7074	0.6989	0.7702	42
Edema	Occlusion top-20%	0.6396	0.7019	0.7671	42
Edema	Occlusion top-30%	0.6192	0.6622	0.7576	42
Pleural Effusion	Baseline Fusion	<u>0.6813</u>	0.7632	0.8542	64
Pleural Effusion	Baseline Fusion top-10%	0.6471	0.7459	0.8552	64
Pleural Effusion	Baseline Fusion top-20%	0.6557	0.7594	<u>0.8784</u>	64
Pleural Effusion	Baseline Fusion top-30%	0.6731	0.7535	**0.8828**	64
Pleural Effusion	Composite	0.6658	<u>0.7656</u>	0.8479	64
Pleural Effusion	GradCAM++	0.5123	0.6744	0.7955	64
Pleural Effusion	GradSHAP	0.6196	0.6737	0.7989	64
Pleural Effusion	GradSHAP top-10%	0.6082	0.7012	0.8058	64
Pleural Effusion	GradSHAP top-20%	0.6082	0.6677	0.7987	64
Pleural Effusion	GradSHAP top-30%	0.6196	0.6687	0.7925	64
Pleural Effusion	Mean Top-1	**0.6873**	**0.7672**	0.8524	64
Pleural Effusion	Mean Top-1^2	0.6663	0.7551	0.8552	64
Pleural Effusion	Mean Top-1^2 top-10%	0.6486	0.7389	0.8491	64
Pleural Effusion	Mean Top-1^2 top-20%	0.6498	0.7404	0.8784	64
Pleural Effusion	Occlusion	0.5725	0.6804	0.8206	64
Pleural Effusion	Occlusion top-10%	0.5644	0.6604	0.7740	64
Pleural Effusion	Occlusion top-20%	0.5967	0.6713	0.7685	64
Pleural Effusion	Occlusion top-30%	0.5919	0.6805	0.8108	64

6.3.5 Generalization to the CheXlocalize Radiologist Test Set

To assess generalization, we repeat the localization evaluation on the held-out CheXlocalize radiologist test set using the same explainability pipelines and fusion strategies. Table 6.12 reports normalized GT mass capture at Top-k, and Table 6.13 reports Top-k dominant-zone retrieval accuracy.

Overall, the ordering of methods is consistent with the validation split: fusion strategies outperform single saliency methods. In particular, Mean Top-1^2 achieves the strongest normalized mass capture (0.8049/0.8511/0.9205 for Top-1/2/3), improving over GradSHAP (0.7431/0.7971/0.8670) and GradCAM++ (0.5795/0.6332/0.8168). Similarly, Mean Top-1^2 yields the best Top-1 zone accuracy (0.5766), while a mildly sparsified variant (Mean Top-1^2 top-20%) performs best at Top-2/Top-3 (0.8161/0.9272), suggesting that moderate sparsification can improve multi-zone retrieval without sacrificing overall fusion quality.

Finally, Table 6.14 reports dominant laterality accuracy on the test set . On the clinically unilateral subset ($N = 149$), Mean Top-1 and Mean Top-1^2 achieve the highest accuracy (0.8658). On the all-cases setting with GT mass-based laterality ($N = 291$), Mean Top-1^2 remains best (0.7457), reinforcing that pathology-aware fusion improves laterality grounding beyond individual

saliency methods.

Table 6.11: Ground-truth pathology mass captured by the model's top-K ranked lung zones.

Experiment	Top-1	Top-2	Top-3
GradSHAP	0.418300	0.666000	0.814600
GradSHAP top-10%	0.405600	0.658500	0.800800
GradSHAP top-20%	0.417200	0.662100	0.807700
GradSHAP top-30%	0.417700	0.661400	0.810000
GradCAM++	0.326200	0.529000	0.767500
Occlusion	0.350300	0.557700	0.716600
Occlusion top-10%	0.358200	0.571700	0.734100
Occlusion top-20%	0.352300	0.571400	0.730500
Occlusion top-30%	0.355300	0.567600	0.726400
Baseline Fusion	0.434100	0.693900	0.845800
Baseline Fusion top-10%	0.420100	0.673900	0.827700
Baseline Fusion top-20%	0.426800	0.683100	0.840000
Baseline Fusion top-30%	0.431100	0.689600	0.847400
Composite	0.440800	0.699300	0.859400
Mean Top-1	0.443400	<u>0.708900</u>	<u>0.862200</u>
Mean Top-1²	**0.453100**	**0.711100**	**0.864900**
Mean Top-1² top-10%	0.438800	0.697200	0.847700
Mean Top-1² top-20%	<u>0.445800</u>	0.704900	0.860100

Table 6.12: Ground-truth mass capture normalized by the maximum achievable GT mass at each Top-K level.

Experiment	Top-1	Top-2	Top-3
GradSHAP	0.743100	0.797100	0.867000
GradSHAP top-10%	0.720500	0.788200	0.852300
GradSHAP top-20%	0.741200	0.792500	0.859600
GradSHAP top-30%	0.742000	0.791600	0.862100
GradCAM++	0.579500	0.633200	0.816800
Occlusion	0.622300	0.667500	0.762700
Occlusion top-10%	0.636300	0.684300	0.781300
Occlusion top-20%	0.625900	0.683900	0.777500
Occlusion top-30%	0.631200	0.679400	0.773100
Baseline Fusion	0.771200	0.830500	0.900200
Baseline Fusion top-10%	0.746300	0.806600	0.880900
Baseline Fusion top-20%	0.758200	0.817600	0.894000
Baseline Fusion top-30%	0.765800	0.825400	0.901900
Composite	0.783100	0.837000	0.914600
Mean Top-1	0.787700	0.848500	0.917600
Mean Top-1²	**0.804900**	**0.851100**	**0.920500**
Mean Top-1² top-10%	0.779500	0.834500	0.902200
Mean Top-1² top-20%	0.792000	0.843700	0.915400

Table 6.13: Top-K zone accuracy across localization strategies on the test set.

Experiment	Top-1	Top-2	Top-3
GradSHAP	0.498100	0.752900	0.887000
GradSHAP top-10%	0.473200	0.733700	0.865900
GradSHAP top-20%	0.488500	0.743300	0.877400
GradSHAP top-30%	0.492300	0.747100	0.879300
GradCAM++	0.369700	0.544100	0.808400
Occlusion	0.410000	0.605400	0.764400
Occlusion top-10%	0.429100	0.616900	0.783500
Occlusion top-20%	0.410000	0.611100	0.781600
Occlusion top-30%	0.411900	0.614900	0.779700
Baseline Fusion	0.557500	0.781600	0.885100
Baseline Fusion top-10%	0.530700	0.764400	0.888900
Baseline Fusion top-20%	0.551700	0.781600	0.894600
Baseline Fusion top-30%	0.547900	0.779700	0.904200
Composite	0.561300	0.791200	0.910000
Mean Top-1	0.561300	0.798900	0.908000
Mean Top-1²	**0.576600**	0.808400	0.925300
Mean Top-1² top-10%	0.557500	0.793100	0.913800
Mean Top-1² top-20%	0.567000	**0.816100**	**0.927200**

Table 6.14: Dominant laterality accuracy for unilateral and bilateral cases on the test set(pathologies : Atelectasis, Consolidation, Pleural Effusion)

Experiment	Acc (unilateral, GT_{inst})	Acc (all, GT_{mass})
GradSHAP	0.8456	0.7388
GradSHAP top-10%	0.8188	0.7182
GradSHAP top-20%	0.8456	0.7320
GradSHAP top-30%	0.8456	0.7354
GradCAM++	0.7181	0.6460
Occlusion	0.7785	0.6529
Occlusion top-10%	0.7919	0.6598
Occlusion top-20%	0.7852	0.6667
Occlusion top-30%	0.7919	0.6770
Baseline Fusion	0.8591	0.7320
Baseline Fusion top-10%	0.8255	0.6942
Baseline Fusion top-20%	0.8523	0.7182
Baseline Fusion top-30%	0.8456	0.7182
Composite	0.8591	0.7320
Mean Top-1	**0.8658**	0.7354
Mean Top-1^2	**0.8658**	**0.7457**
Mean Top-1^2 top-10%	**0.8658**	0.7285
Mean Top-1^2 top-20%	**0.8658**	<u>0.7423</u>

6.4 Decision-zone validation for report generation

Because report language is gated by pathology-specific decision zones, we first validate that the learned thresholds produce clinically meaningful operating regions. For each pathology, a capability profile is derived to obtain T_{gate} (rule-out), $T_{\text{intersection}}$ (intermediate operating point) and $T_{\text{rule-in}}$ (high-specificity rule-in). Thresholds are derived on the CheXpert train/validation split (non-radiologist validation) using calibrated probabilities, subject to Spec ≥ 0.92, Sens ≥ 0.94, and TP ≥ 30 to avoid unstable estimates on rare findings. The resulting thresholds are then held fixed for decision-zone assignment and report gating in the validation set and the test set. Zone-wise behavior is summarized using posterior risk and likelihood ratios (Zone 0: LR^-; Zone 1: intermediate evidence; Zone 2: LR^+).

Table 6.15: Pathology-specific decision-zone thresholds derived on the CheXpert validation split. T_{gate} defines the rule-out gate, $T_{\text{intersection}}$ the intermediate operating point, and $T_{\text{rule-in}}$ the high-specificity rule-in threshold.

Pathology	$\mathbf{T}_{\text{gate}}$	$\mathbf{T}_{\text{intersection}}$	$\mathbf{T}_{\text{rule-in}}$
Atelectasis	0.082	0.367	0.702
Cardiomegaly	0.079	0.348	0.754
Consolidation	0.010	0.085	0.272
Edema	0.279	0.577	0.829
Pleural Effusion	0.334	0.557	0.774

Figure 6.2 shows an increase in posterior disease probability across the decision zones, supporting the intended semantics of rule-out (Zone 0), intermediate/uncertain region (Zone 1a), and rule-in (Zone 1b/2). In the validation split, Zone 0 is associated with very low posterior risk, whereas Zone 2 concentrates the highest-risk cases, providing a principled basis for suppressing findings in Zone 0 and using more assertive language in Zone 2. Figure 6.3 further indicates that Zone 2 yields moderate-to-large diagnostic strength (LR^+) across multiple findings, while Zone 1b generally corresponds to smaller-to-moderate evidence. Together, these results justify the employment of this decision zone strategy for report generation where conservative phrasing is used for the intermediate zone and stronger phrasing used only when evidence falls in a high-specificity rule-in region. Also, Figure 6.4 summarizes the diagnostic strength associated with each decision zone using likelihood ratios on a log scale specifically on the radiologist test set.

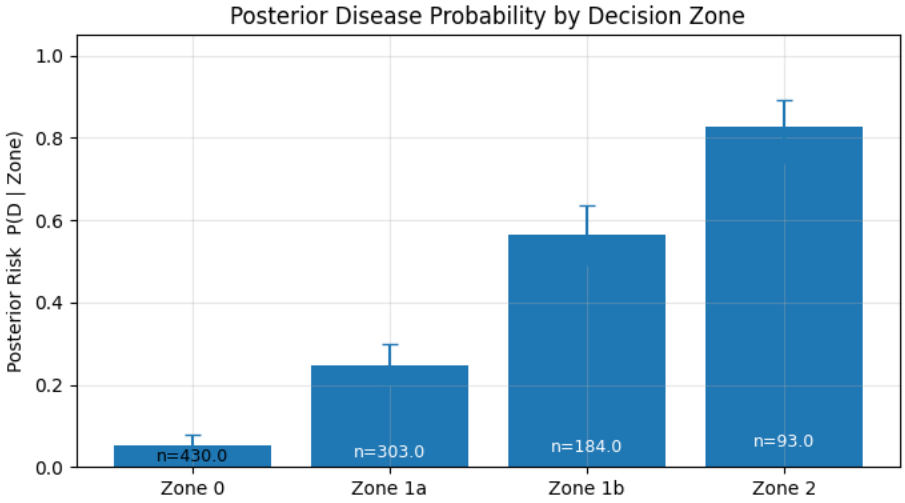


Figure 6.2: Posterior disease probability by decision zone on the CheXpert validation split. Error bars indicate uncertainty; n denotes the number of cases assigned to each zone.

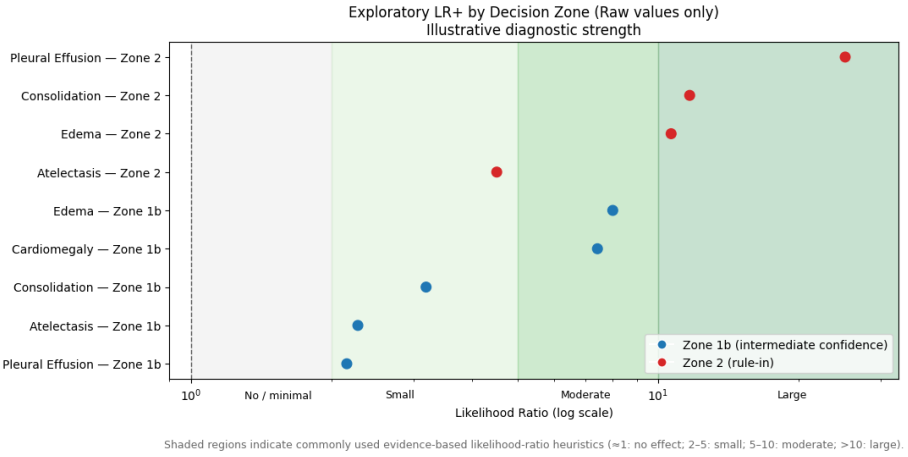


Figure 6.3: Exploratory diagnostic strength by decision zone, shown as likelihood ratio (log scaled) on the validation set. Shaded bands indicate common LR interpretation ranges (approximately: 2–5 small, 5–10 moderate, > 10 large).

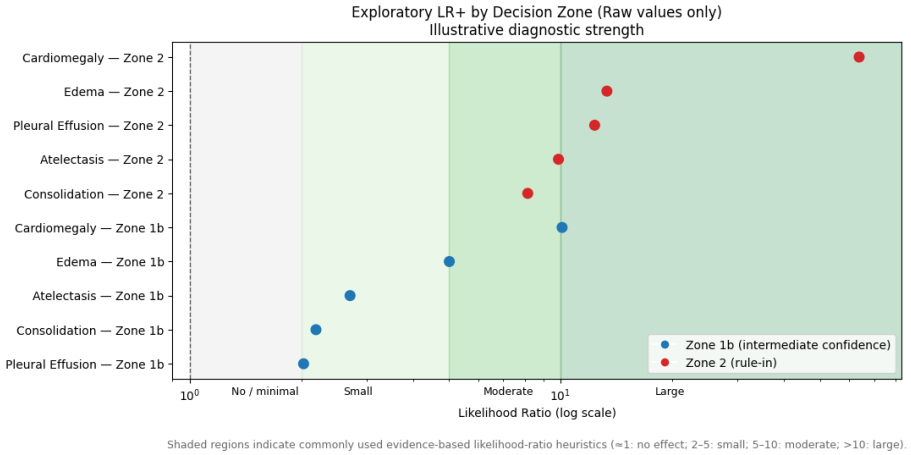


Figure 6.4: Diagnostic strength by decision zone shown as likelihood ratio(log scaled) on the test set. Shaded bands indicate common LR interpretation ranges (approximately: 2–5 small, 5–10 moderate, > 10 large).

6.5 Radiology report evaluation

This subsection evaluates whether the generated reports are faithful to the structured evidence provided to the language model. Report evaluation is performed on the radiologist reference reports available for the same validation cohort used in the localization analysis ($N = 200$ studies). For each study, the generator receives only structured, positive evidence derived from the classifier and the explainability pipeline: pathology presence, decision-zone–derived confidence band, and (for lung findings) a small Top-k set of coarse anatomical zones. The specific Large Language model that was selected as our radiology report generator is the gpt-4o-mini model.

In this thesis we do not report F1-RadGraph scores. Although RadGraph-based metrics quantify overlap of extracted entities and relations between reference and generated reports, they are primarily designed for evaluating a comprehensive report generation. In contrast, our framework intentionally produces constrained, pathology-scoped text conditioned on structured evidence and decision-zone gating, and it explicitly avoids describing non-target findings or incidental observations. As a result, a whole-report graph-overlap score would largely reflect divergence in reporting scope and style (e.g., omission of devices, secondary findings, or unrelated anatomic relations) rather than the clinical objectives of this work, namely label fidelity for the five target pathologies, adherence to decision-aware language constraints, and consistency with zone-level evidence. We therefore prioritize evaluation protocols that directly measure these goals, including pathology-level agreement (CheXbert) and quantitative localization/laterality metrics grounded in CheXlocalize.

6.5.1 CheXbert evaluation

We quantify report-level fidelity using CheXbert by comparing labels extracted from the generated report against labels extracted from the radiologist reference report, restricted to the target pathology set. Per-pathology precision, recall, and F1 are reported, alongside macro-averaged scores across the evaluated pathologies.

Table 6.16: Per-pathology CheXbert agreement metrics for synthetic reports (case-level).

Pathology	Accuracy	Precision	Recall	F1	TP	FP	FN	TN
Atelectasis	0.420	0.238	0.897	0.376	35	112	4	49
Cardiomegaly	0.605	0.235	0.852	0.368	23	75	4	98
Consolidation	0.490	0.099	0.846	0.177	11	100	2	87
Edema	0.680	0.377	0.644	0.475	29	48	16	107
Pleural Effusion	0.750	0.567	0.873	0.688	55	42	8	95

At the study level, CheXbert agreement reveals a clear precision–recall trade-off across pathologies. Pleural effusion achieves the strongest overall performance, with the highest F1 score (0.688) and the most balanced precision–recall profile, followed by edema (F1 = 0.475). In contrast, atelectasis, cardiomegaly, and especially consolidation exhibit high recall but substantially lower precision, indicating that the generated reports tend to over-mention these findings relative to the reference reports. This pattern suggests that the constrained generation setup remains sensitive to supported abnormalities, but still produces pathology-specific false-positive wording for findings with broader or more ambiguous imaging manifestations. To contextualize the per-pathology precision/recall trade-offs in Table 6.16, we additionally inspected the raw CheXbert outputs at the case level (positive = 1, uncertain = -1, negative = 0, and not-mentioned/undetected = NaN) for the same evaluation subset ($N = 200$ studies; 1000 pathology–case pairs). Because the primary metrics follow a strict, literature-aligned mapping (only 1 counts as positive; $\{0, -1, \text{NaN}\}$ are treated as non-positive), false negatives (FNs) can arise either from missing mentions in the generated report (gen = NaN) or from uncertainty phrasing that CheXbert assigns as -1 and is therefore penalized as negative under strict scoring. FN mechanisms differed by pathology: atelectasis, cardiomegaly, and pleural effusion were dominated by omission-driven errors, whereas edema and consolidation more often reflected uncertainty-marked wording in the generated text. False negatives (FNs) can arise either from missing mentions in the generated report (gen = NaN) or from uncertainty phrasing that CheXbert assigns as -1 and is therefore penalized as negative under strict scoring. FN mechanisms differed by pathology: atelectasis, cardiomegaly, and pleural effusion were dominated by omission-driven errors, whereas edema and consolidation more often

reflected uncertainty-marked wording in the generated text.Crucially, an analysis of the False Positives (FPs) revealed a significant metric limitation: CheXbert frequently misclassifies safely hedged, low-confidence language generated from Zone 1a (e.g., indeterminate", cannot exclude") as hard positive assertions (1.0). As detailed in Table 6.17, these indeterminate Zone 1a predictions account for the majority of the total False Positives across all evaluated pathologies, ranging from 53.6% for atelectasis up to 81.2% for edema. While true over-calling and explicit negation conflicts (ref = 0) do occur—particularly for cardiomegaly and consolidation—this metric artifact artificially inflates the FP rate and disproportionately penalizes the framework for expressing appropriately calibrated uncertainty. Overall, this decomposition suggests that CheXbert disagreements heavily reflect this rigid handling of uncertainty, rather than pure model hallucination.

Pathology	**Zone 1a Share of Total FPs**
Atelectasis	53.6%
Cardiomegaly	77.3%
Consolidation	66.0%
Edema	81.2%
Pleural Effusion	61.9%

Table 6.17: Proportion of CheXbert False Positives (FPs) originating from generated reports restricted to Zone 1a (indeterminate) language.

Case A: Uncertainty-driven FN (Edema) `patient64611_study1_view1_frontal`
CheXbert (ref/gen): $1/-1$ **Mechanism:** strict scoring penalizes uncertain mentions
Ref: Indistinctness of the pulmonary vasculature consistent with pulmonary edema.
Gen: Probable pulmonary edema is observed, involving the right and left mid and upper lung zones.

Case 1: Edema (Explicit Negation Penalty)
CheXbert (ref/gen): 0/1 **Mechanism:** metric penalizes safely hedged uncertainty against explicit negation
Ref: There is no evidence for edema.
Gen: ... Edema is present in multiple lung zones, but the findings are **indeterminate**.

Case 2: Consolidation (Explicit Negation Penalty)
CheXbert (ref/gen): 0/1 **Mechanism:** failure to parse LLM exclusion warnings
Ref: ... no focal consolidation, and no significant pulmonary edema.
Gen: Consolidation, edema, and pleural effusion **cannot be excluded** based on the current findings.

Case 3: Pleural Effusion (Explicit Negation Penalty)
CheXbert (ref/gen): 0/1 **Mechanism:** strict binarization of equivocal phrasing
Ref: There is no evidence of pleural effusion.
Gen: ... Pleural effusion is **indeterminate**, with potential involvement in the left lower lung, right lower lung, and left mid lung.

Case 4: Atelectasis (Uncertainty Penalty)
CheXbert (ref/gen): −1/1 **Mechanism:** reference uncertainty (-1) conflicts with generated uncertainty (1)
Ref: ... redemonstration of small linear opacity in the retrocardiac region, which **may represent** some atelectasis.
Gen: Indeterminate atelectasis is noted in the left lower lung and left mid lung zones.

Case 5: Cardiomegaly (Uncertainty Penalty)
CheXbert (ref/gen): −1/1 **Mechanism:** reference uncertainty (-1) conflicts with generated uncertainty (1)
Ref: Cardiac silhouette and vascularity are **mildly prominent**.
Gen: Cardiomegaly is noted; however, the finding is **indeterminate**.

Figure 6.5: Representative cases illustrating CheXbert False Positives (FPs) originating from generated reports restricted to Zone 1a.

To complement the disagreement analysis, Figure 6.6 provides representative agreement cases under the same strict CheXbert mapping (positive = 1; non-positive $\in \{0, -1, \text{NaN}\}$). In the true-positive examples, both the reference and generated reports explicitly verbalize the same target finding (e.g., pulmonary edema or pleural effusion), yielding consistent positive CheXbert assignments (1/1). These cases illustrate that, when evidence is included and phrased with canonical pathology terms, the constrained generator can produce clinically plausible statements that align with the reference report at the label level. Conversely, the true-negative example demonstrates conservative behavior: when the reference does not mention pleural effusion, the generated report likewise avoids introducing the finding, resulting in matched non-positive assignments (0/0). Together, these agreement cases indicate that the framework can both (i) correctly surface salient abnormalities and (ii) remain appropriately silent when there is no supported evidence to report, consistent with the positive-only reporting policy.

Case D: True Positive (Edema) `patient64557_study1_view1_frontal`
CheXbert (ref/gen): 1/1 **Interpretation:** consistent positive mention
Ref: Persistent mild-to-moderate pulmonary edema, grossly unchanged.
Gen: Conclusive evidence of edema in the left mid/lower lung and right mid lung regions.

Case E: True Negative (Pleural Effusion) `patient64651_study1_view1_frontal`
CheXbert (ref/gen): 0/0 **Interpretation:** consistent absence of mention
Ref: No focal consolidation. (No pleural effusion described.)
Gen: No acute cardiopulmonary abnormality is noted.

Case F: True Positive (Pleural Effusion) `patient64655_study1_view1_frontal`
CheXbert (ref/gen): 1/1 **Interpretation:** consistent positive mention
Ref: Left retrocardiac opacity and a small left pleural effusion, unchanged from the prior exam.
Gen: Suggestive pleural effusion involving the left mid/lower lung regions (and mid lung zones bilaterally).

Figure 6.6: Representative CheXbert agreement (success) cases illustrating correct positive and negative mention alignment under strict scoring.

To further illustrate the practical value of this uncertainty-aware design, Figure 6.7 presents a sensitivity analysis comparing strict CheXbert scoring against the true clinical utility of the framework.

Observing this analysis yields two critical clinical insights:

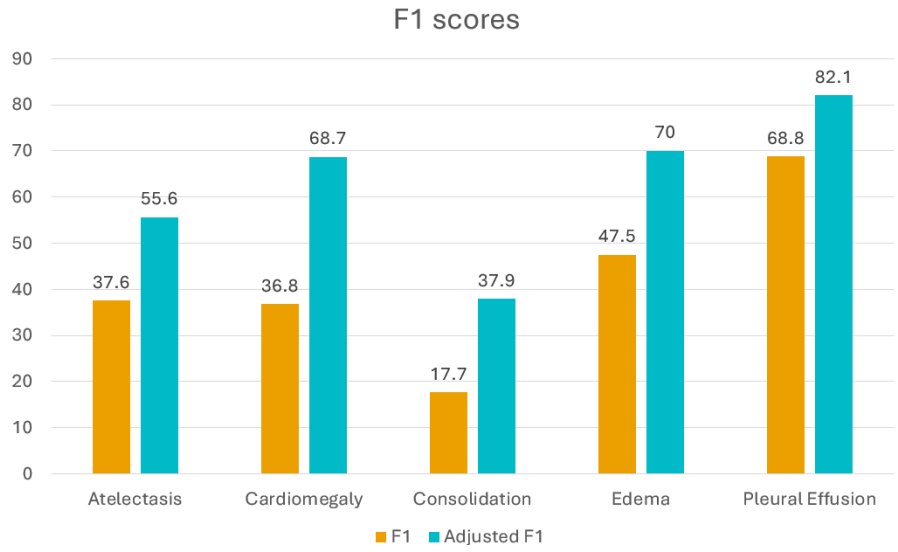


Figure 6.7: This plot demonstrates the effective sensitivity of the framework when safely hedged Zone 1a predictions are treated as clinically actionable warnings rather than strict False Positives.

- **Reclaiming Sensitivity through Safety:** Under strict NLP metrics, an "indeterminate" generated statement is penalized. However, in a clinical setting, successfully flagging a borderline radiograph as "cannot exclude" is a diagnostic success. The analysis shows that when Zone 1a cases are properly recognized as safety-nets rather than errors, the effective sensitivity of the system increases dramatically across all pathologies.
- **Mitigating Automation Bias:** Binary classifiers force a definitive "yes" or "no," which can mislead clinicians on ambiguous scans. By capturing these borderline probabilities in Zone 1a, the framework trades a slight drop in automated precision for a massive gain in clinical safety, effectively preventing dangerous False Negatives (missed diagnoses) without committing to hallucinated certainties.

6.6 Summary of Evaluation Findings

This chapter evaluated the proposed decision-aware interpretability pipeline at three levels: (i) zone-level localization against radiologist polygons (Section 6.3), (ii) decision-zone validity for report gating (Section 6.4), and (iii) report-level fidelity and safety under constrained generation (Section 6.5).

6.6.1 Key Localization Findings

Across the CheXlocalize radiologist validation instances ($N = 279$), fusion improves spatial grounding over any single saliency method. In particular, pathology-aware fusion with an elitist weighting rule (Mean Top-1^2) provides the strongest overall alignment, achieving normalized GT coverage of 0.662/0.757/0.859 at Top-1/Top-2/Top-3, and the best Top-k zone accuracy (0.480/0.756/0.867). Among single-method baselines, GradSHAP is consistently stronger than Grad-CAM++ and Occlusion, while sparsification (top-p%) shows mixed effects—sometimes helping at higher k, but often reducing strict Top-1 performance.

Per-pathology analysis highlights clinically plausible heterogeneity: compact targets such as cardiomegaly are easier to localize than diffuse entities such as edema, motivating Top-2/Top-3 reporting and zone-based normalization. Laterality-related metrics should be interpreted in light of the underlying GT laterality distribution; for example, edema is predominantly bilateral in this split, which limits the clinical usefulness of strict left–right discrimination for that finding.

6.6.2 Key Report-Level Findings

Report evaluation (Section 6.5) focuses on whether the generated text remains faithful to the structured evidence and avoids clinically unsafe hallucinations. Overall, the constrained reporting setup is designed so that: (i) ruled-out findings are explicitly suppressed, (ii) uncertain cases are expressed with calibrated hedging language, and (iii) rule-in cases are reported more assertively, while still restricting anatomical claims to the coarse lung-zone evidence.

Automatic label-fidelity metrics (CheXbert agreement on target pathologies) provide a tractable measure of whether the report preserves the intended positive/negative status under the imposed constraints, while hallucination/violation audits quantify safety failures (e.g., mentioning forbidden pathologies or adding unsupported detail). Where included, pathology-aware sentence-level evaluation provides additional clinical grounding by testing whether the report's pathology-specific sentences align with the intended anatomy and laterality cues, rather than only matching global wording.

Chapter 7

Discussion

This chapter discusses the findings from the evaluation results and the implications that arise from them. We primarily focus on the efficacy of zone-based explainability—employed via individual or fused saliency methods—and the success of the clinical decision-zone framework in guiding safe radiology report generation.

7.1 Summary of Findings

Regarding the localization evaluation on the CheXlocalize dataset, the pathology-aware weighted combination of saliency methods significantly improved localization compared to using single methods individually. Because thoracic findings are highly diffuse—with only 56% of an abnormality's mass typically concentrated in the single most dominant region —evaluating retrieval based on the maximum achievable ground-truth mass provides the clearest picture of performance. Under this metric, our best configuration (the squared $MeanTop - 1^2$ variant) successfully retrieved 80% of the maximum possible abnormality using just the Top-1 predicted zone, and 85% using the Top-2 zones.The evaluation on the CheXlocalize radiologist-annotated test set yielded consistent findings. The fused experiments maintained the strongest results, proving that zone-based explainability generalizes effectively to unseen data.Analysis of the decision-zone modeling further validates our approach. The calculated likelihood ratios (LR+) across the predicted zones exhibited a monotonic pattern that perfectly maps to Jaeschke-style evidence-strength bands, clinically justifying our use of distinct zones to constrain LLM vocabulary. Specifically, a finding falling into Zone 0 (Rule-out) is omitted; Zone 1a (Indeterminate) triggers highly hedged uncertainty phrasing; Zone 1b (Suggestive) uses cautious reporting language; and Zone 2 (Rule-in) permits conclusive, assertive phrasing. Finally, the CheXbert evaluation of the generated reports revealed satisfactory recall results and also a limitation of the metric itself, which does not recognize correctly the language used for the zone 1a as it labels it as hard positives very frequently.

7.2 Interpretability and Clinical Plausibility

Radiologists routinely localize findings to standard lung regions (e.g., "right lower lung") when dictating reports. Therefore, projecting continuous saliency heatmaps and radiologist polygons onto a discrete 6-zone grid provides a highly intuitive, clinically aligned interface. Furthermore, the ground-truth data confirms that abnormalities frequently span across at least two of the six zones , reinforcing the clinical necessity of our Top-k retrieval evaluation and the generation of multi-zone textual descriptions.Another major finding is the success of the pathology-aware fusion strategy. By utilizing a weighted average based on validation accuracy—and specifically by applying a squaring operation to create an "elitist" filter—the framework effectively exponentially rewards the most reliable method for a specific pathology while suppressing noisy, failing methods. This proves that different thoracic findings benefit from different explainability mechanisms.

7.3 Limitations

An important limitation of this study is that the evaluation of the generated radiology reports was constrained to the CheXpert validation set, as the reference reports corresponding to the CheXlocalize test set are not publicly available. A further limitation concerns the pathology-specific α-relative saliency threshold used to dynamically select the zones passed to the report generator. This adaptive rule was chosen to balance evidence compactness against spatial recall. Consequently, some highly diffuse abnormalities might be represented by fewer zones than would be captured under a fixed Top-3 policy, potentially understating the full spatial extent of the disease. Conversely, a fixed policy risks injecting irrelevant regions for highly focal findings. The α-based rule is a pragmatic compromise, rather than an exact estimator of ground-truth disease boundaries.Additionally, while the 6-zone lung system is highly interpretable, it lacks the granularity to represent precise anatomical boundaries. The generated text is strictly constrained to avoid fine-grained anatomical claims to prevent hallucinations. Furthermore, radiographs are occasionally not centrally aligned with the mediastinum; lateral shifts can cause the fixed geometric grid to inaccurately reflect the true anatomical distribution of the disease. Finally, the automated CheXbert evaluation metric presents a distinct limitation regarding uncertainty. CheXbert strictly binarizes findings, which heavily penalizes clinically appropriate uncertainty language. Our analysis showed that a vast majority of the recorded False Positives (e.g., 81.2% for Edema, 77.3% for Cardiomegaly) were simply instances where CheXbert misclassified our safely hedged Zone 1a language (e.g., "indeterminate") as hard positive assertions. Thus, while CheXbert is a useful baseline, it is insufficient as a standalone measure of clinical safety and utility.

7.4 Future Work

Future iterations of this framework should prioritize human expert evaluation of the generated reports to assess whether the decision-zone phrasing and localization statements match clinical expectations beyond rigid NLP label agreement. Specifically, deploying segmentation models to identify the spine and lung fields would provide reliable anatomical landmarks to automatically "straighten" the image, ensuring proper horizontal alignment and rotation to 0 degrees before the grid is applied. Also, a direct comparison between the α-based dynamic selection rule and fixed Top-k prompting strategies would also help quantify the optimal trade-off between spatial coverage and textual compactness. Finally, the framework could be scaled to process the remaining CheXpert pathologies (e.g., lung lesion, lung opacity, pneumonia, pneumothorax, fracture) to provide comprehensive diagnostic coverage.

7.5 Conclusion

Overall, this thesis demonstrates a viable pathway for elevating XAI from qualitative visual heatmaps into calibrated, decision-aware evidence that drives safe, uncertainty-aware report generation. The primary insight is that clinically meaningful interpretability requires intermediate structure: anatomically grounded zones for spatial reasoning, and statistically derived decision zones for calibrated communication. The remaining challenges—incorporating richer anatomical mapping, broadening the reporting scope, and conducting robust clinical evaluations—define clear directions for future research toward trustworthy, automated clinical deployment.

Appendix A

Appendices

A.1 YAML Configuration Files

Listing A.1: Training configuration for the ViT-L/14 CLIP-based multi-label chest X-ray classifier.

```
# vitl_clip336_freeze8_proguf_a100.yaml

exp_name: vitl_clip336_freeze8_proguf_a100

model: vit_large_patch14_clip_336.openai
num_classes: 13
pretrained: True

work_dir: /content/drive/MyDrive/chexpert_runs
data_root: /content
train_csv: /content/train_full_r1.csv
val_csv: /content/val_full_r1.csv
holdout_csv: /content/valid.csv

# Loss
loss: asl
asl_gamma_pos: 1
asl_gamma_neg: 4

# Optimization
optimizer: adamw
lr: 3e-5
weight_decay: 0.02
scheduler: cosine
warmup_epochs: 1
```

```yaml
# Architecture / Fine-tuning
freeze_first_blocks: 8
progressive_unfreeze: True
dropout: 0.1
epochs: 6
batch_size: 48
image_size: 336

# Mixup and Augmentations
mixup_alpha: 0.1
cutmix_alpha: 0.0

# Precision
use_amp: True
amp_dtype: bfloat16

# Logging
use_wandb: True
wandb_project: chexpert_experiments
save_best_metric: val_auc
save_top_k: 1

# Misc
accumulation_steps: 1
early_stopping_patience: 4
```

A.2 Per-Class Metrics Tables

Table A.1: Per-class discrimination performance on the validation set (CheXpert).

Class	**ROC-AUC**	**PR-AUC**
Atelectasis	0.788	0.599
Cardiomegaly	0.788	0.696
Consolidation	0.906	0.558
Edema	0.921	0.713
Enlarged Cardiomediastinum	0.502	0.493
Lung Lesion	0.659	0.012
Lung Opacity	0.918	0.922
Pleural Effusion	0.917	0.806
Pleural Other	0.970	0.125
Pneumonia	0.658	0.186
Pneumothorax	0.809	0.435
Support Devices	0.953	0.940

Table A.2: Per-class discrimination performance on the test set (CheXpert).

Class	**ROC-AUC**	**PR-AUC**
Atelectasis	0.823	0.601
Cardiomegaly	0.890	0.776
Consolidation	0.875	0.310
Edema	0.893	0.524
Pleural Effusion	0.951	0.840
Enlarged Cardiomediastinum	0.594	0.576
Lung Opacity	0.911	0.896
Lung Lesion	0.863	0.314
Pneumonia	0.790	0.224
Pneumothorax	0.991	0.683
Pleural Other	0.982	0.420
Fracture	0.570	0.065
Support Devices	0.967	0.957

A.3 Clinical decision zones per pathology (tuned on val set from the train/val split)

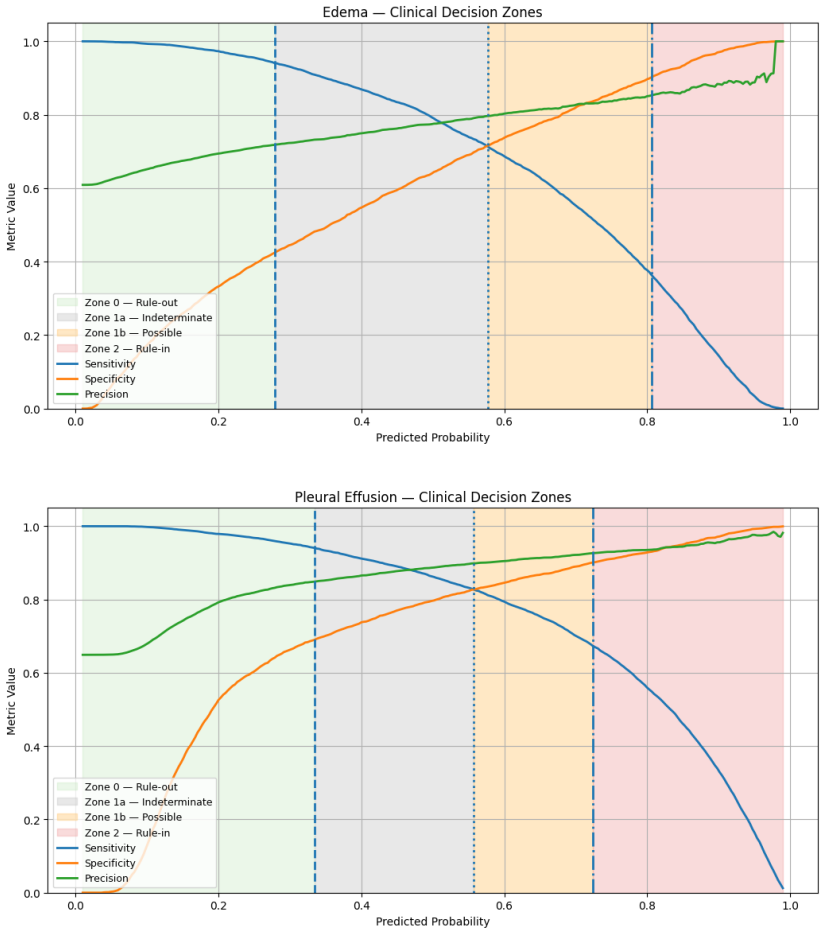


Figure A.1: Clinical decision zones induced by calibrated probability thresholds. (Part 1 of 3: Edema and Pleural Effusion)

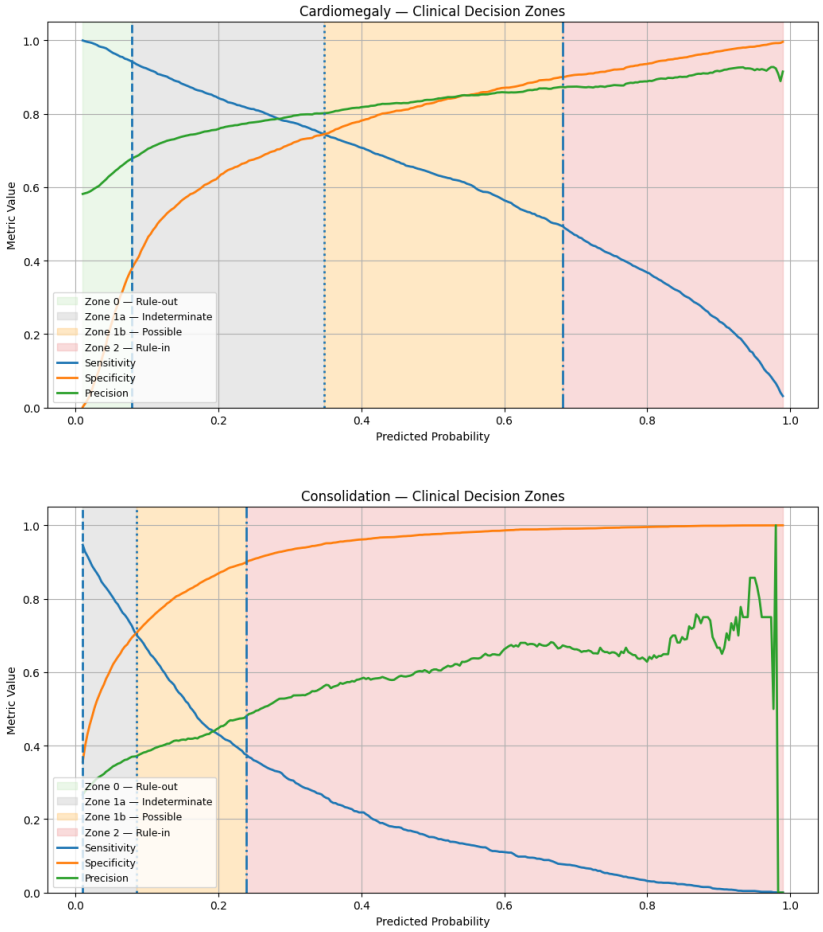


Figure A.1: Clinical decision zones induced by calibrated probability thresholds. (Part 2 of 3: Cardiomegaly and Consolidation)

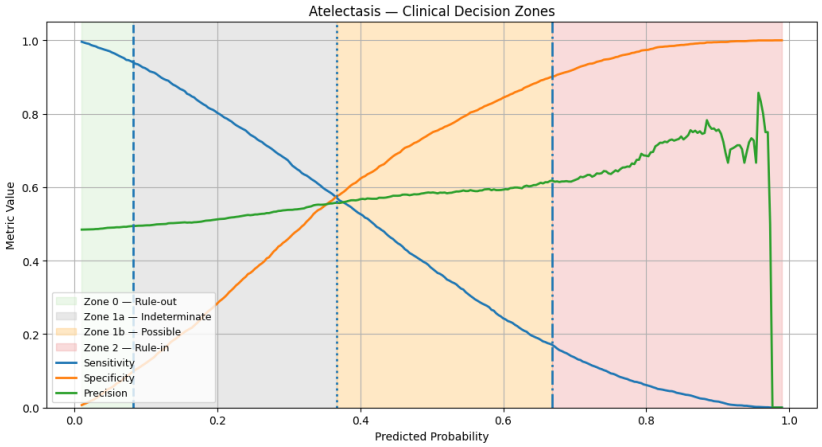


Figure A.1: Clinical decision zones induced by calibrated probability thresholds. (Part 3 of 3: Atelectasis). Vertical markers indicate the derived thresholds T_{gate}, $T_{\text{intersection}}$, and $T_{\text{rule-in}}$, partitioning the probability axis into rule-out (Zone 0), indeterminate (Zone 1a), possible (Zone 1b), and rule-in (Zone 2) regions.

A.4 Representative CheXbert disagreement cases

Case A (Uncertainty-driven FN): patient64611_study1_view1_frontal (Edema; ref= 1, gen= −1)

Reference text :

> AP view of the supine chest shows right upper extremity subclavian venous catheter with tip in the atriocaval junction. Lung volumes are low. Indistinctness of the pulmonary vasculature consistent with pulmonary edema.

Generated text :

> There is probable edema noted in the right mid lung, right upper lung, and left mid lung regions. Probable pulmonary edema is observed, with involvement of the right and left mid and upper lung zones.

Case B (Hard contradiction FP): `patient64573_study1_view1_frontal` (Edema; ref= 0, gen= 1)

Reference text :

> The lung volumes remain low, with mild degree of bibasilar opacities, likely atelectasis. No focal consolidation, and no significant pulmonary edema.

Generated text :

> There is evidence of edema in the right mid lung, left upper lung, and left mid lung regions. The confidence level for these findings is indeterminate.

Case C (Omission-driven FN): `patient64624_study1_view1_frontal` (Pleural Effusion; ref= 1, gen= NaN)

Reference text :

There is blunting of the left costophrenic angle which may represent a small pleural effusion. ... Small left pleural effusion.

Generated text :

There is probable consolidation in the left mid and lower lung zones. Cardiomegaly is suggestive. Atelectasis cannot be excluded in the left mid and lower lung zones.

Bibliography

[1] H. Q. Nguyen, K. Lam, L. T. Le, H. H. Pham, D. Q. Tran, D. B. Nguyen, D. D. Le, C. M. Pham, H. T. Tong, D. H. Dinh et al., "Vindr-cxr: An open dataset of chest x-rays with radiologist's annotations," Scientific Data, vol. 9, no. 1, p. 429, 2022.

[2] J. Irvin, P. Rajpurkar, M. Ko, Y. Yu, S. Ciurea-Ilcus, C. Chute, H. Marklund, B. Haghgoo, R. Ball, K. Shpanskaya et al., "Chexpert: A large chest radiograph dataset with uncertainty labels and expert comparison," in Proceedings of the AAAI conference on artificial intelligence, vol. 33, no. 01, 2019, pp. 590–597.

[3] S. Park, G. Kim, Y. Oh, J. B. Seo, S. M. Lee, J. H. Kim, S. Moon, J.-K. Lim, C. M. Park, and J. C. Ye, "Self-evolving vision transformer for chest x-ray diagnosis through knowledge distillation," Nature communications, vol. 13, no. 1, p. 3848, 2022.

[4] S. Rajaraman, P. Ganesan, and S. Antani, "Deep learning model calibration for improving performance in class-imbalanced medical image classification tasks," PloS one, vol. 17, no. 1, p. e0262838, 2022.

[5] A. Saporta, X. Gui, A. Agrawal, A. Pareek, S. Q. Truong, C. D. Nguyen, V.-D. Ngo, J. Seekins, F. G. Blankenberg, A. Y. Ng et al., "Benchmarking saliency methods for chest x-ray interpretation," Nature Machine Intelligence, vol. 4, no. 10, pp. 867–878, 2022.

[6] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, "Sanity checks for saliency maps," Advances in neural information processing systems, vol. 31, 2018.

[7] J. Hatherley, R. Sparrow, and M. Howard, "The virtues of interpretable medical ai," Cambridge Quarterly of Healthcare Ethics, vol. 33, no. 3, pp. 323–332, 2024.

[8] S. Rahimiaghdam and H. Alemdar, "Evaluating the quality of visual explanations on chest x-ray images for thorax diseases classification," Neural Computing and Applications, vol. 36, no. 17, pp. 10 239–10 255, 2024.

[9] A. White, K. H. Ngan, J. Phelan, K. Ryan, S. S. Afgeh, C. C. Reyes-Aldasoro, and A. d. Garcez, "Contrastive counterfactual visual explanations with overdetermination," Machine Learning, vol. 112, no. 9, pp. 3497–3525, 2023.

[10] J. Kim, S. Kwak, H. Lee, J. Chang, and S. M. Park, "Explainable opportunistic osteoporosis screening from chest x-rays: a retrospective comparison of foundation models," Osteoporosis International, pp. 1–11, 2025.

[11] M. Aasem and M. Javed Iqbal, "Toward explainable ai in radiology: Ensemble-cam for effective thoracic disease localization in chest x-ray images using weak supervised learning," Frontiers in big Data, vol. 7, p. 1366415, 2024.

[12] R. Dwivedi, P. Kothari, D. Chopra, M. Singh, and R. Kumar, "An efficient ensemble explainable ai (xai) approach for morphed face detection," Pattern Recognition Letters, vol. 184, pp. 197–204, 2024.

[13] M. Salvi, S. Seoni, A. Campagner, A. Gertych, U. R. Acharya, F. Molinari, and F. Cabitza, "Explainability and uncertainty: Two sides of the same coin for enhancing the interpretability of deep learning models in healthcare," International Journal of Medical Informatics, vol. 197, p. 105846, 2025.

[14] B. Jing, P. Xie, and E. Xing, "On the automatic generation of medical imaging reports," in Proceedings of the 56th annual meeting of the association for computational linguistics (volume 1: long papers), 2018, pp. 2577–2586.

[15] Y. Xue, T. Xu, L. Rodney Long, Z. Xue, S. Antani, G. R. Thoma, and X. Huang, "Multimodal recurrent model with attention for automated radiology report generation," in International Conference on Medical Image Computing and Computer-Assisted Intervention. Springer, 2018, pp. 457–466.

[16] J. Yuan, H. Liao, R. Luo, and J. Luo, "Automatic radiology report generation based on multi-view image fusion and medical concept enrichment," in International conference on medical image computing and computer-assisted intervention. Springer, 2019, pp. 721–729.

[17] Z. Chen, Y. Song, T.-H. Chang, and X. Wan, "Generating radiology reports via memory-driven transformer," in Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP), 2020, pp. 1439–1449.

[18] M. Rakibul Islam, M. Zahid Hossain, M. Ahmed, and M. Sharmin Sultana Samu, "Vision-language models for automated chest x-ray interpretation: Leveraging vit and gpt-2," Engineering Reports, vol. 7, no. 6, p. e70220, 2025.

[19] F. Yu, M. Endo, R. Krishnan, I. Pan, A. Tsai, E. P. Reis, E. Fonseca, H. Lee, Z. Shakeri, A. Ng et al., "Radiology report expert evaluation (rexval) dataset," 2023.

[20] F. Yu, M. Endo, R. Krishnan, I. Pan, A. Tsai, E. P. Reis, E. K. U. N. Fonseca, H. M. H. Lee, Z. S. H. Abad, A. Y. Ng et al., "Evaluating progress in automatic chest x-ray radiology report generation," Patterns, vol. 4, no. 9, 2023.

[21] A. Smit, S. Jain, P. Rajpurkar, A. Pareek, A. Y. Ng, and M. Lungren, "Combining automatic labelers and expert annotations for accurate radiology report labeling using bert," in Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP), 2020, pp. 1500–1519.

[22] S. Jain, A. Agrawal, A. Saporta, S. Q. Truong, D. N. Duong, T. Bui, P. Chambon, Y. Zhang, M. P. Lungren, A. Y. Ng et al., "Radgraph: Extracting clinical entities and relations from radiology reports," arXiv preprint arXiv:2106.14463, 2021.

[23] B. McCrindle, K. Zukotynski, T. E. Doyle, and M. D. Noseworthy, "A radiology-focused review of predictive uncertainty for ai interpretability in computer-assisted segmentation," Radiology: Artificial Intelligence, vol. 3, no. 6, p. e210031, 2021.

[24] N. Karunanayake, "Confidence in radiology ai: From black-box scores to trusted decisions," European Journal of Radiology Artificial Intelligence, p. 100046, 2025.

[25] R. M. Qafesha, M. D. Hindawi, I. Sharabati, M. Dervis, Q. Najah, A. Albostani, and A. H. G. Ali, "Diagnostic performance of artificial intelligence in radiographs for pneumoperitoneum detection: a systematic review and meta-analysis," British Journal of Radiology, vol. 99, no. 1177, pp. 37–49, 2026.

[26] M. H. Bernstein, M. Chung, A. Yala, and G. L. Baird, "Triage with ai: A rule-out framework quantifying the risks and benefits of screening mammogram automation," medRxiv, pp. 2025–04, 2025.

[27] D. Resch, R. Lo Gullo, J. Teuwen, F. Semturs, J. Hummel, A. Resch, and K. Pinker, "Ai-enhanced mammography with digital breast tomosynthesis for breast cancer detection: Clinical value and comparison with human performance," Radiology: Imaging Cancer, vol. 6, no. 4, p. e230149, 2024.

[28] E. Begoli, T. Bhattacharya, and D. Kusnezov, "The need for uncertainty quantification in machine-assisted medical decision making," Nature Machine Intelligence, vol. 1, no. 1, pp. 20–23, 2019.

[29] R. D. Riley, G. S. Collins, L. Kirton, K. I. Snell, J. Ensor, R. Whittle, P. Dhiman, M. Van Smeden, X. Liu, J. Alderman et al., "Uncertainty of risk estimates from clinical prediction models: rationale, challenges, and approaches," bmj, vol. 388, 2025.

[30] A. B. Shinagare, P. R. Shankar, V. Chernyak, S. A. Woolen, B. R. Herts, E. M. Azene, D. G. Mitchell, A. B. Rosenkrantz, K. Raghavan, B. Karmazyn et al., "Potential frameworks for communicating diagnostic certainty in radiology reports: From the acr commission on quality and safety," Journal of the American College of Radiology, vol. 22, no. 11, pp. 1390–1398, 2025.

[31] Center for Artificial Intelligence in Medicine and Imaging, "Chexlocalize dataset," 2022, dataset. [Online]. Available: https://doi.org/10.71718/hap9-kn94

[32] R. Jaeschke, G. H. Guyatt, D. L. Sackett, G. Guyatt, E. Bass, P. Brill-Edwards, G. Browman, D. Cook, M. Farkouh, H. Gerstein et al., "Users' guides to the medical literature: Iii. how to use an article about a diagnostic test b. what are the results and will they help me in caring for my patients?" Jama, vol. 271, no. 9, pp. 703–707, 1994.